

BFS Traversal

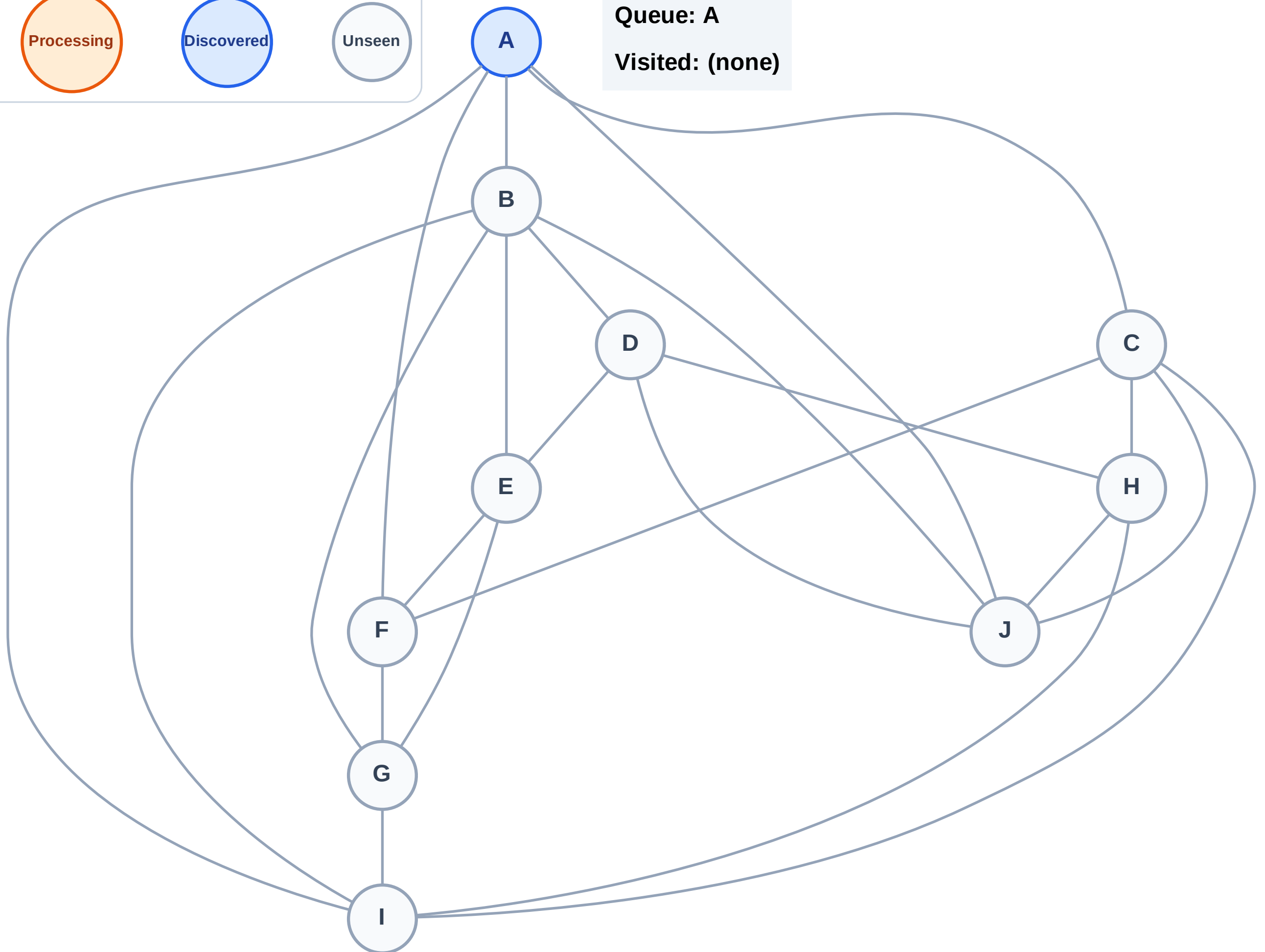
Step 1: Start at A

Legend



Queue: A

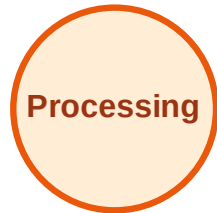
Visited: (none)



BFS Traversal

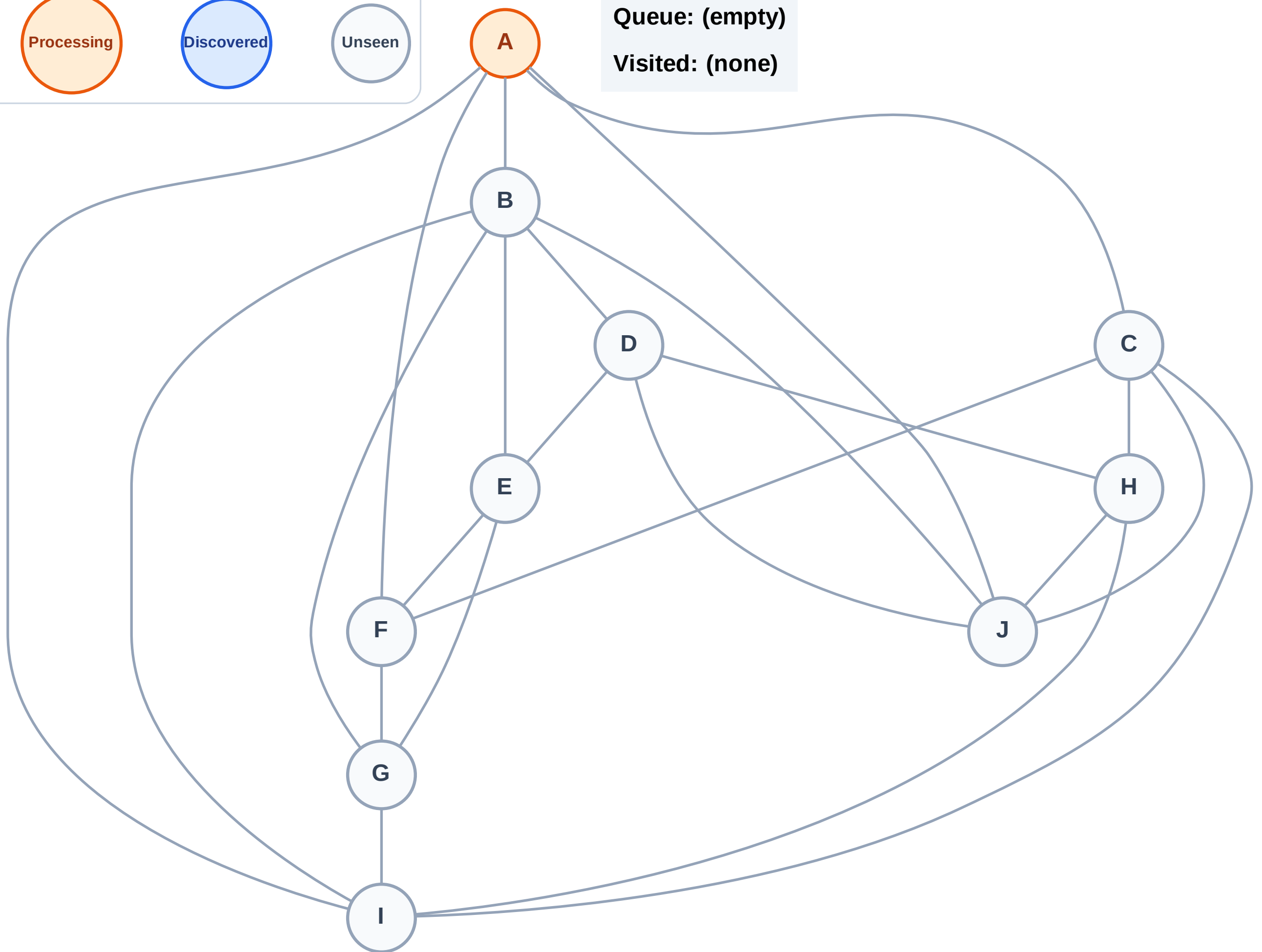
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



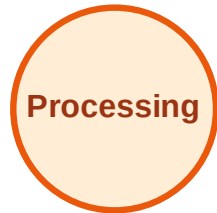
BFS Traversal

Step 3: Discover B, C, F, I, J from A

Legend



Complete



Processing

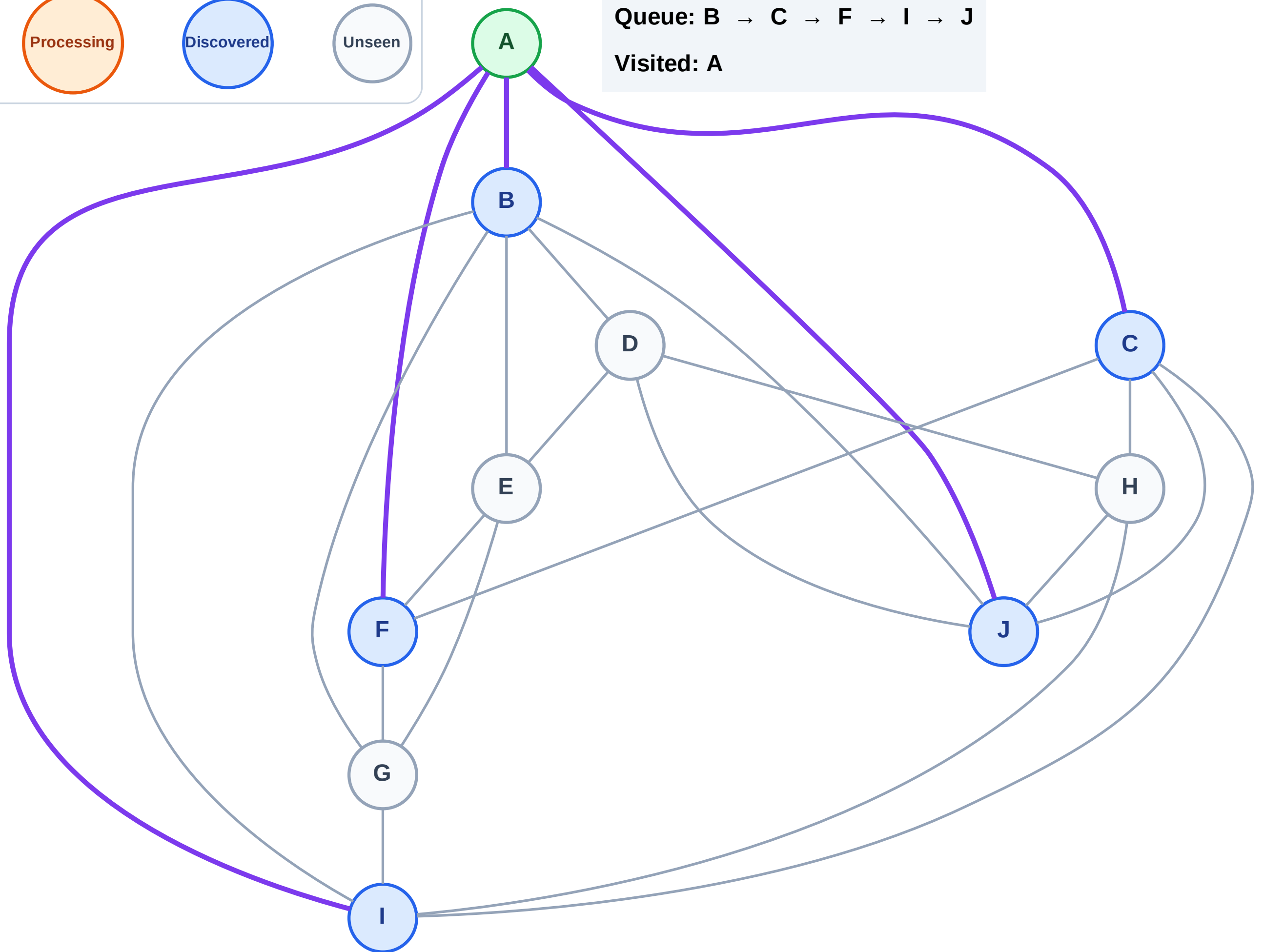


Discovered



Unseen

Queue: B → C → F → I → J
Visited: A



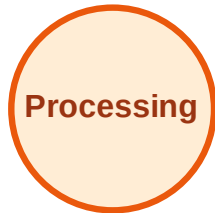
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



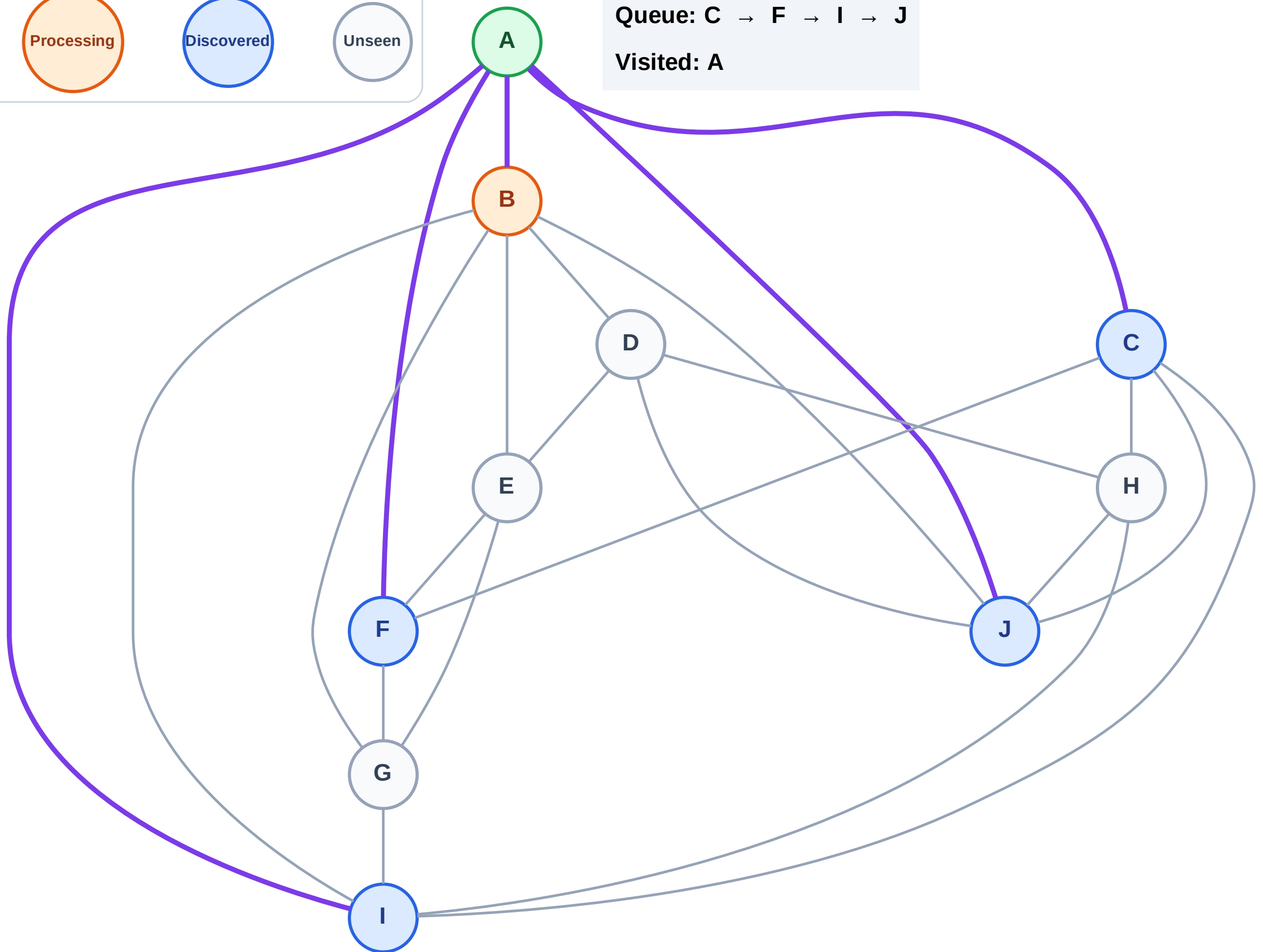
Discovered



Unseen

Queue: C → F → I → J

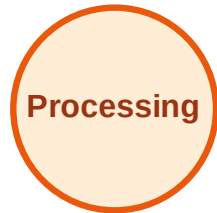
Visited: A



BFS Traversal

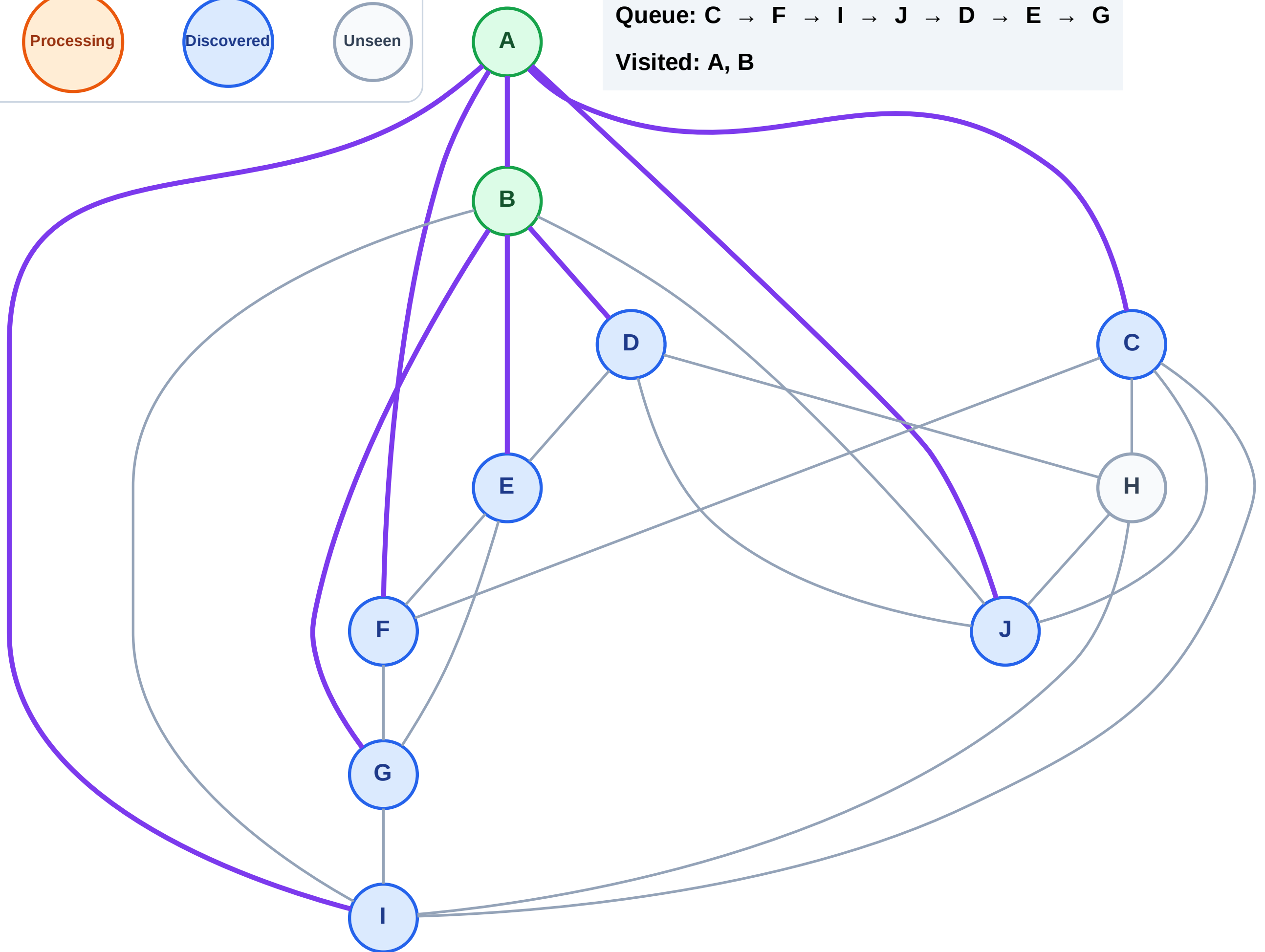
Step 5: Discover D, E, G from B

Legend



Queue: C → F → I → J → D → E → G

Visited: A, B



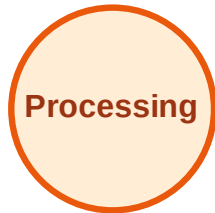
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



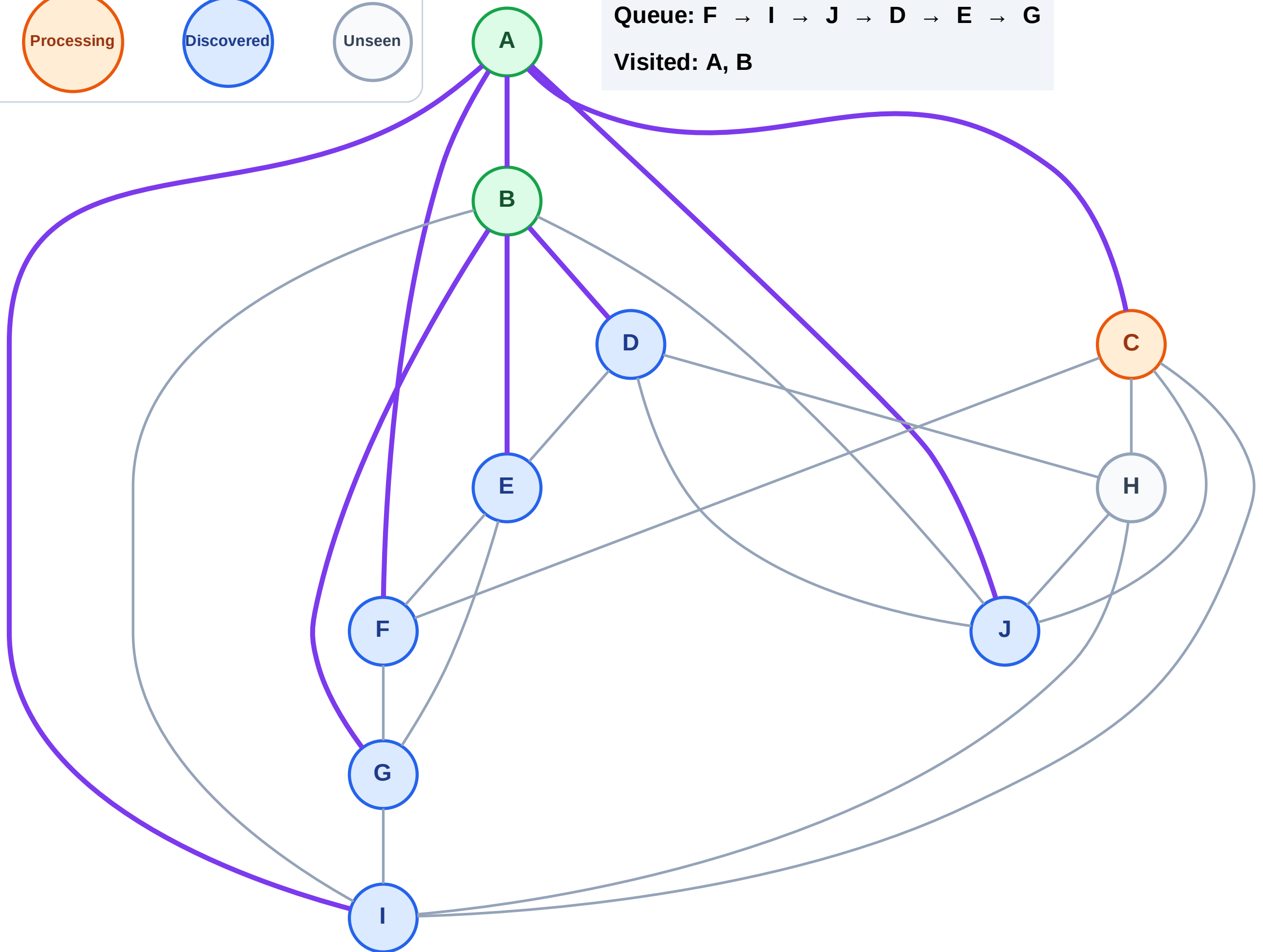
Discovered



Unseen

Queue: F → I → J → D → E → G

Visited: A, B



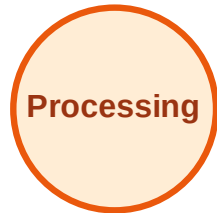
BFS Traversal

Step 7: Discover H from C

Legend



Complete



Processing



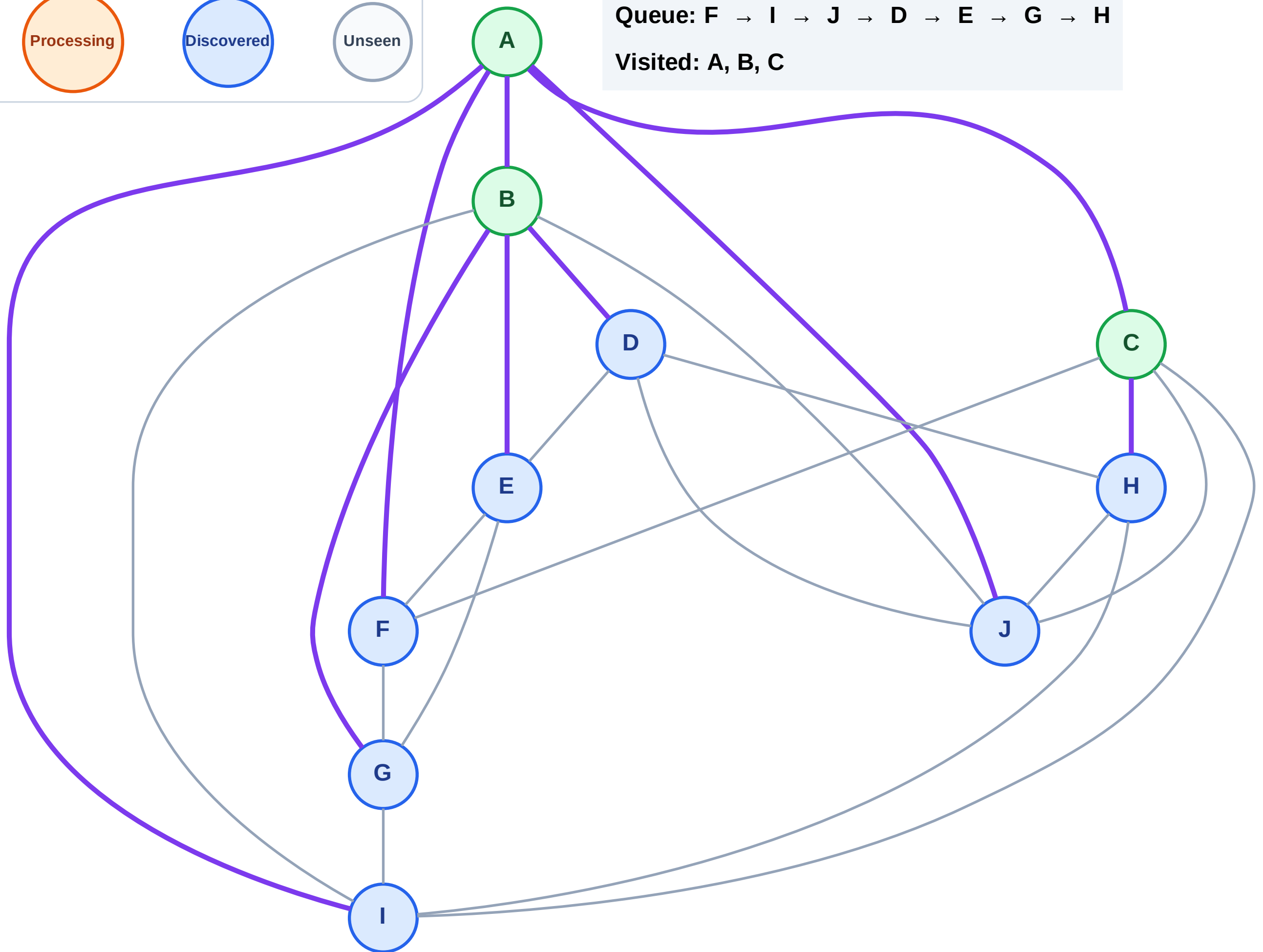
Discovered



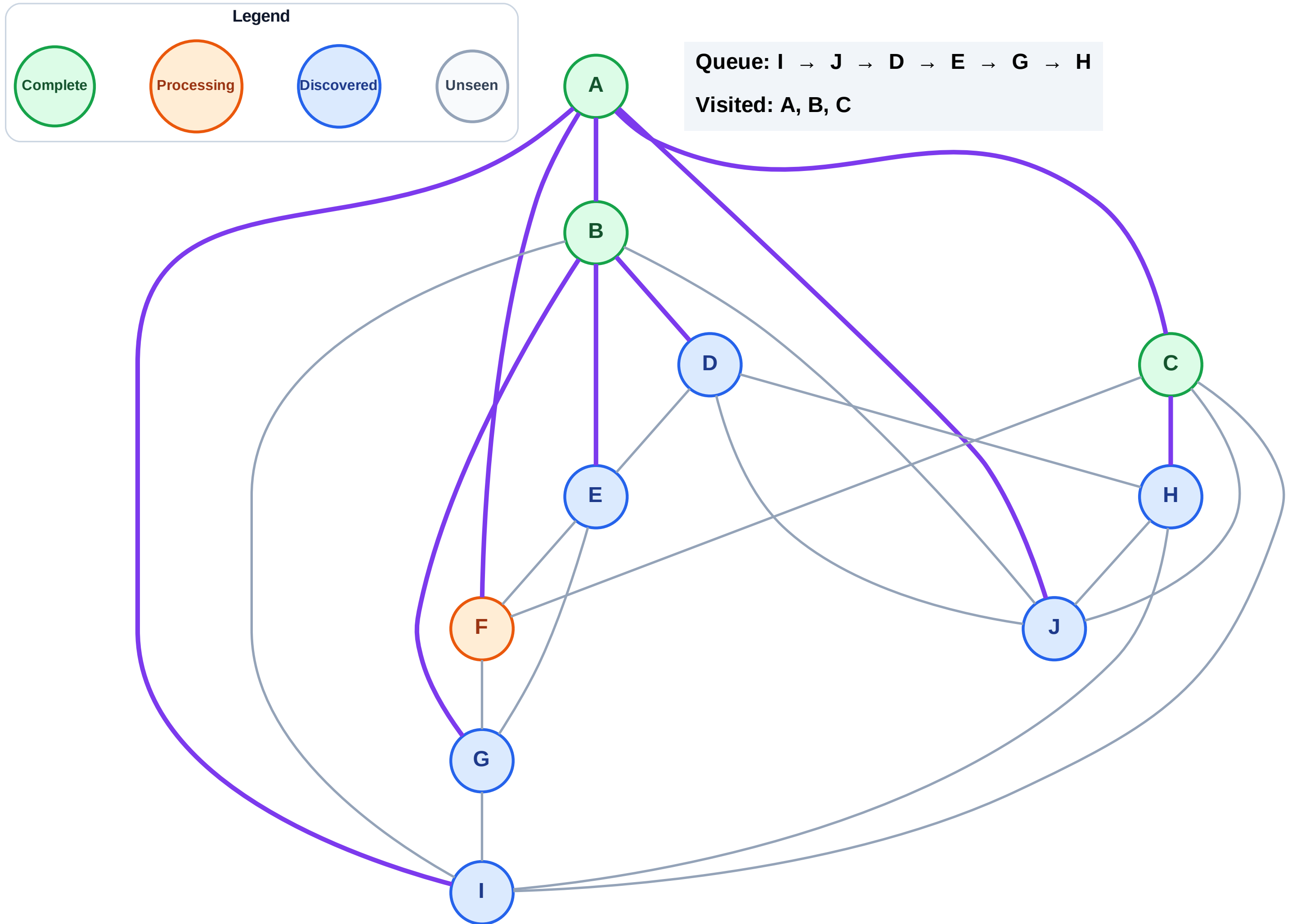
Unseen

Queue: F → I → J → D → E → G → H

Visited: A, B, C



Step 8: Process node F



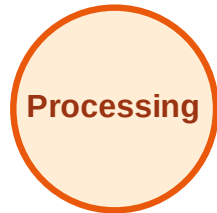
BFS Traversal

Step 9: No new neighbors from F

Legend



Complete



Processing



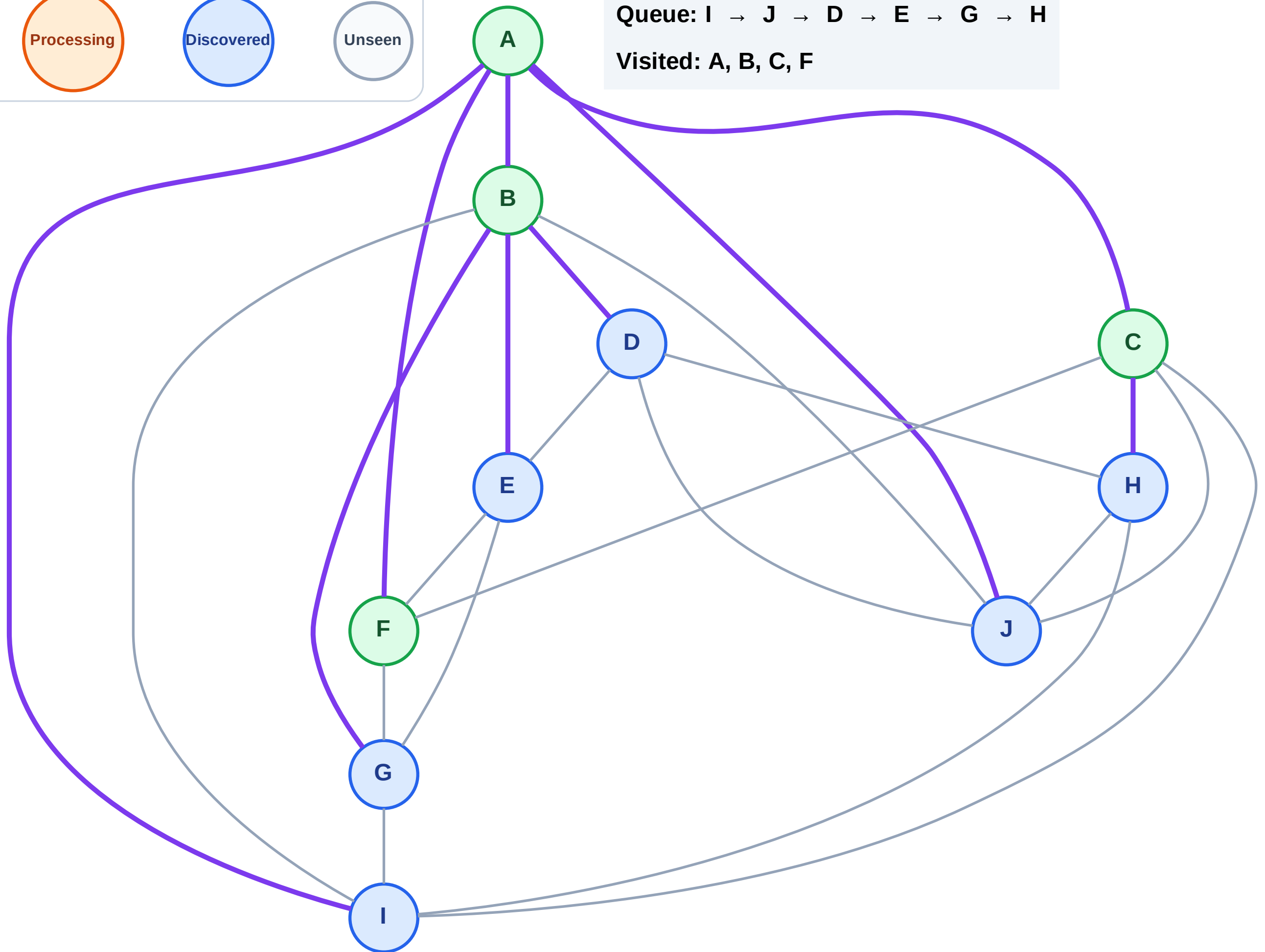
Discovered



Unseen

Queue: I → J → D → E → G → H

Visited: A, B, C, F



BFS Traversal

Step 10: Process node I

Legend

Complete

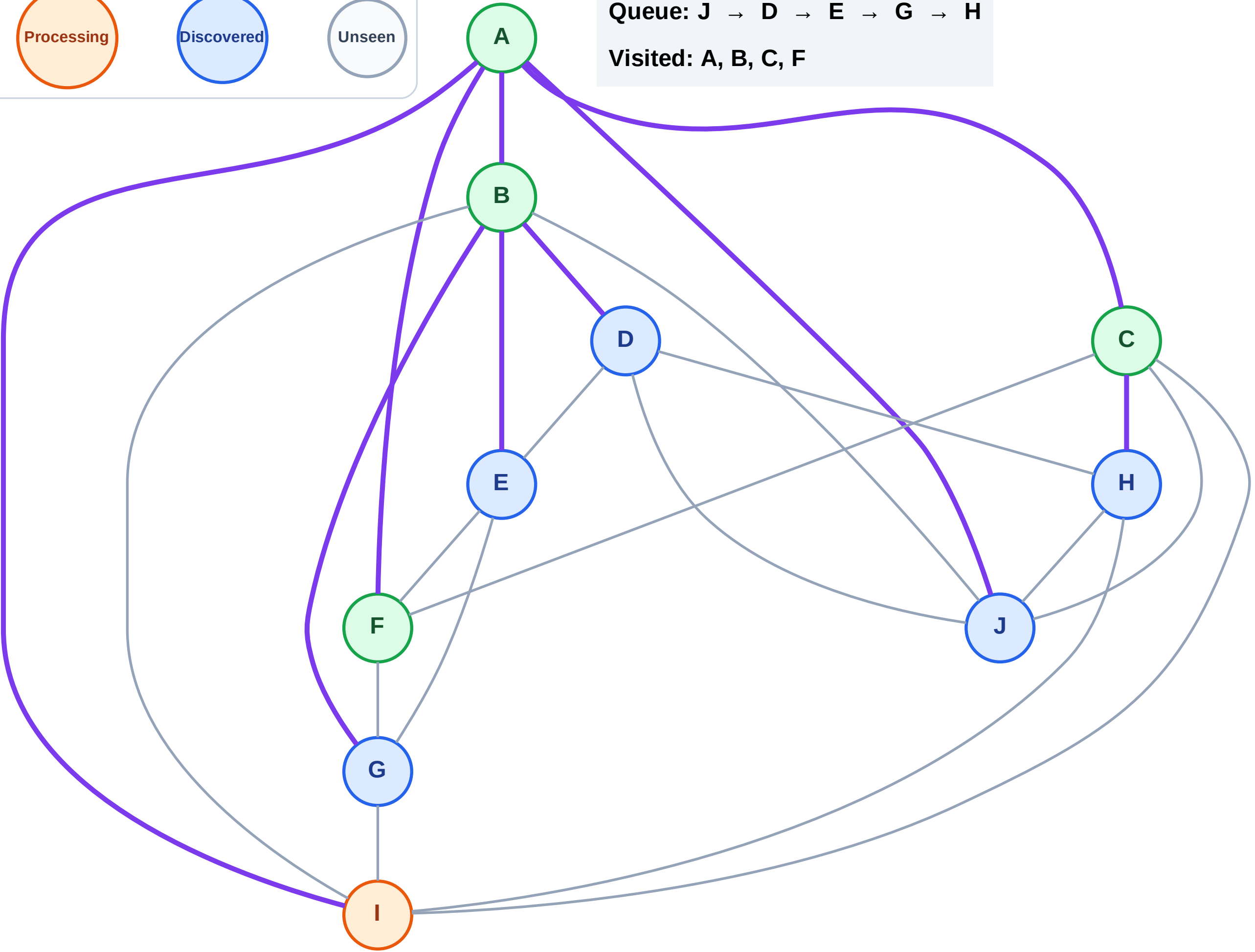
Processing

Discovered

Unseen

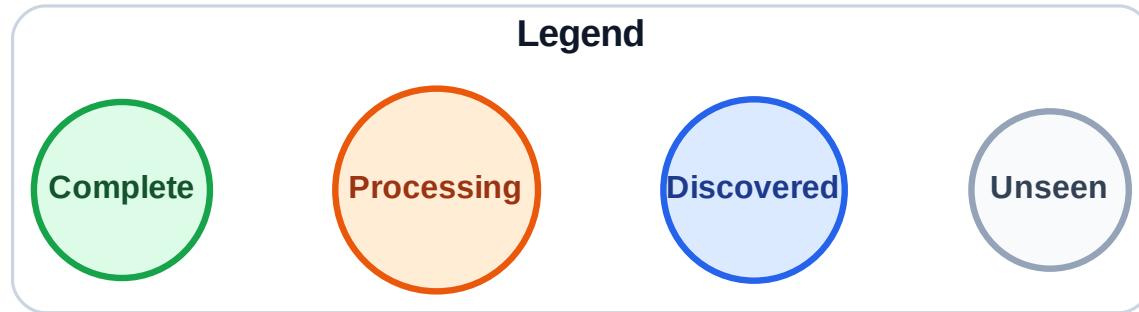
Queue: J → D → E → G → H

Visited: A, B, C, F



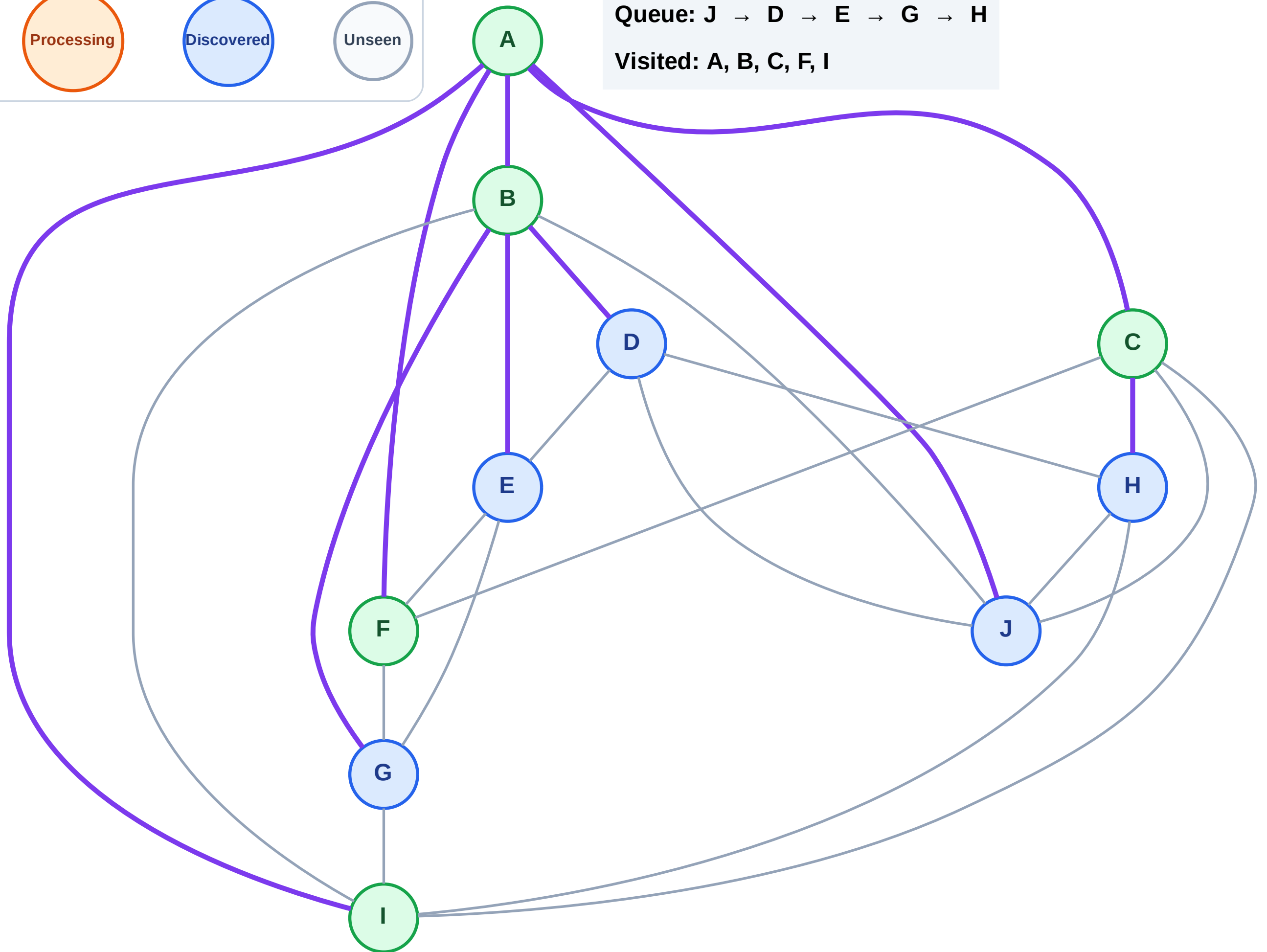
BFS Traversal

Step 11: No new neighbors from I



Queue: J → D → E → G → H

Visited: A, B, C, F, I



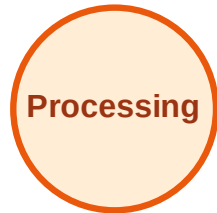
BFS Traversal

Step 12: Process node J

Legend



Complete



Processing



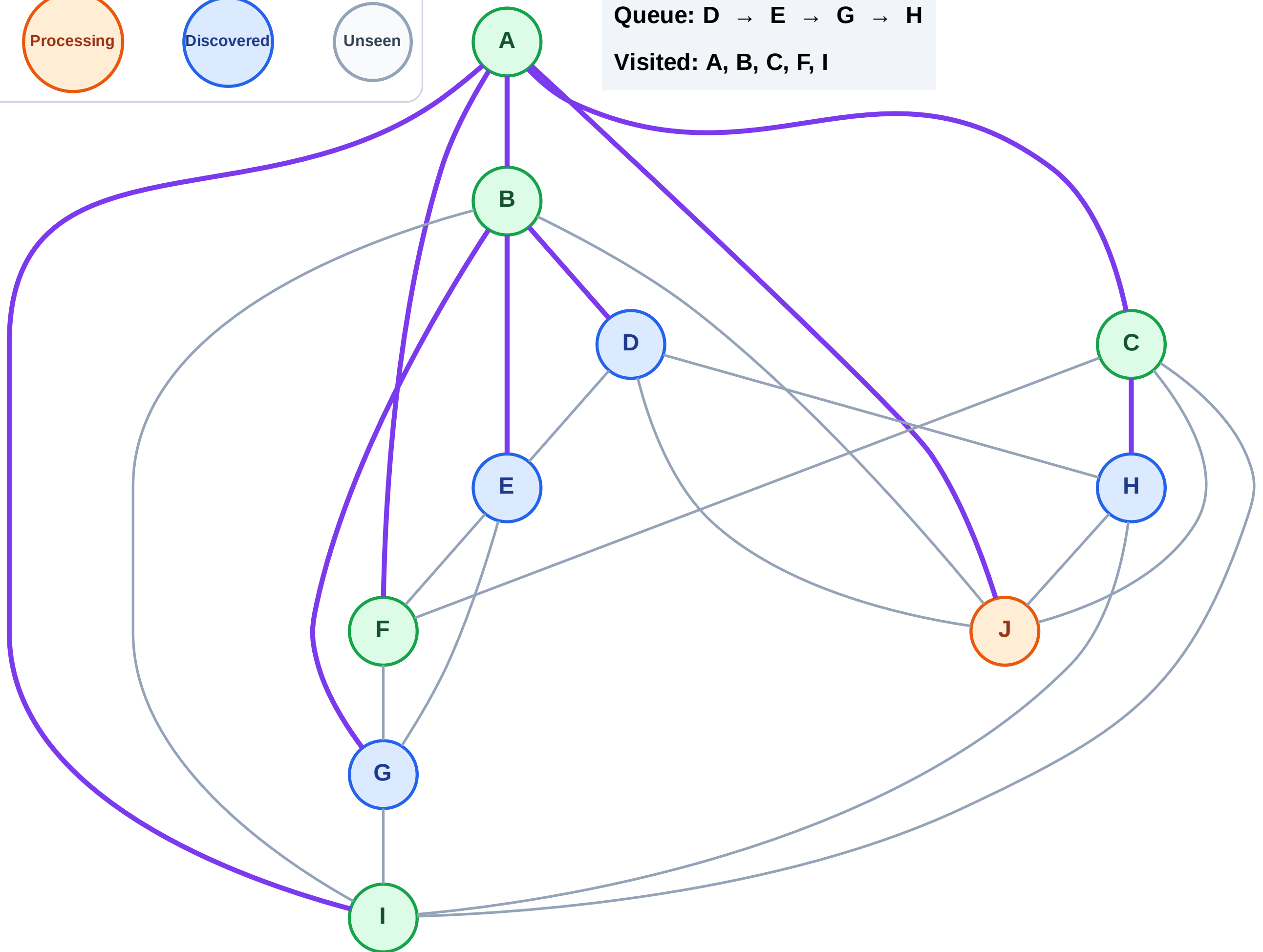
Discovered



Unseen

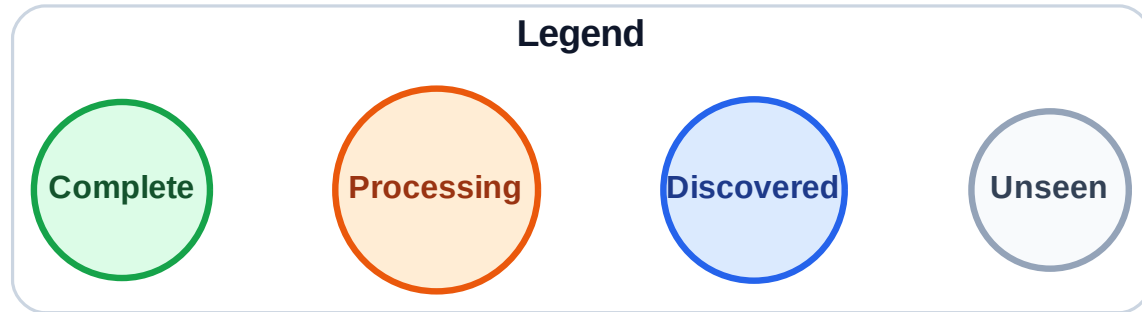
Queue: D → E → G → H

Visited: A, B, C, F, I



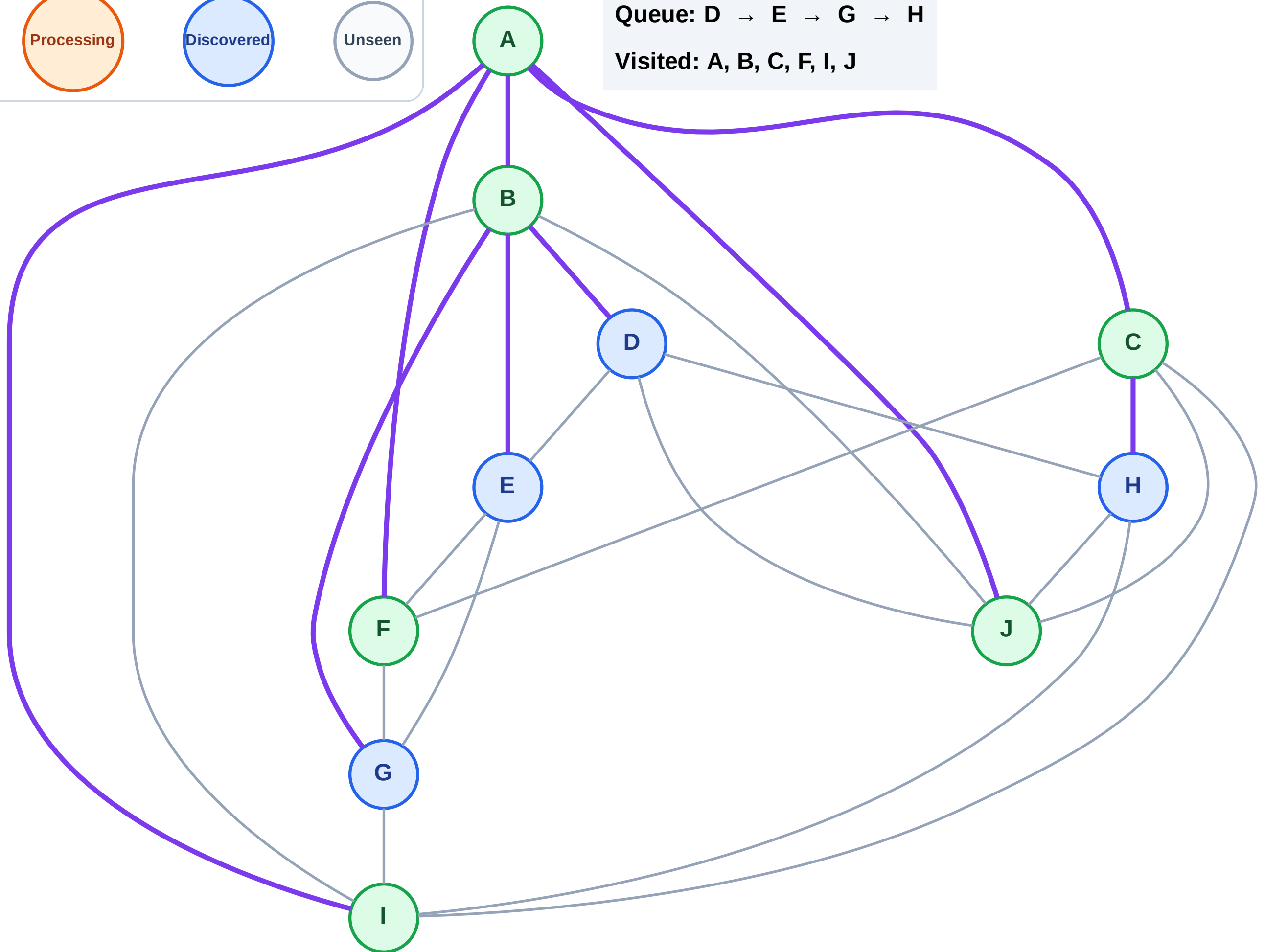
BFS Traversal

Step 13: No new neighbors from J



Queue: D → E → G → H

Visited: A, B, C, F, I, J



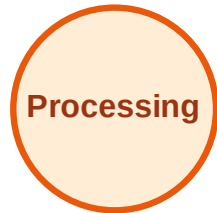
BFS Traversal

Step 14: Process node D

Legend



Complete



Processing



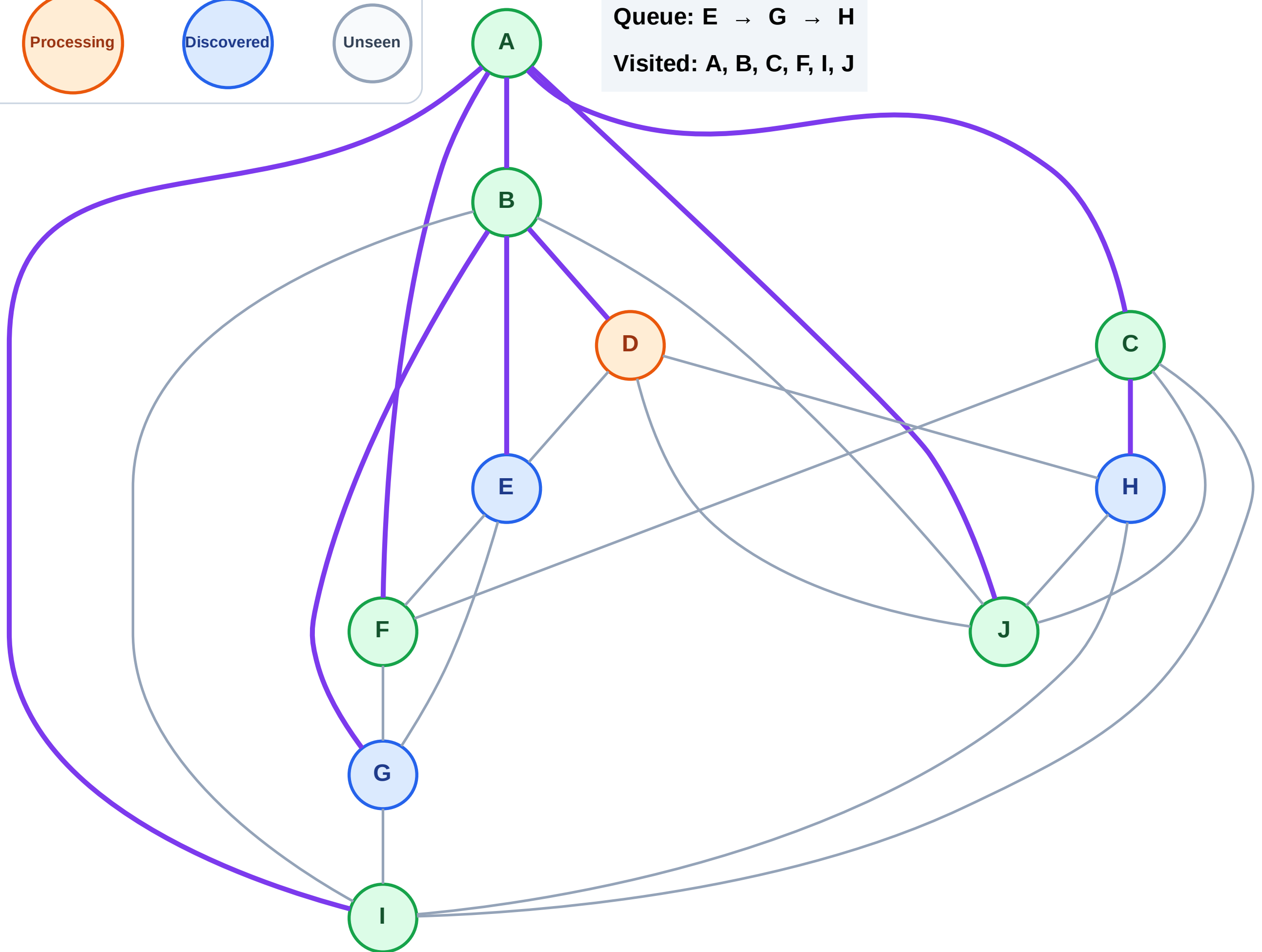
Discovered



Unseen

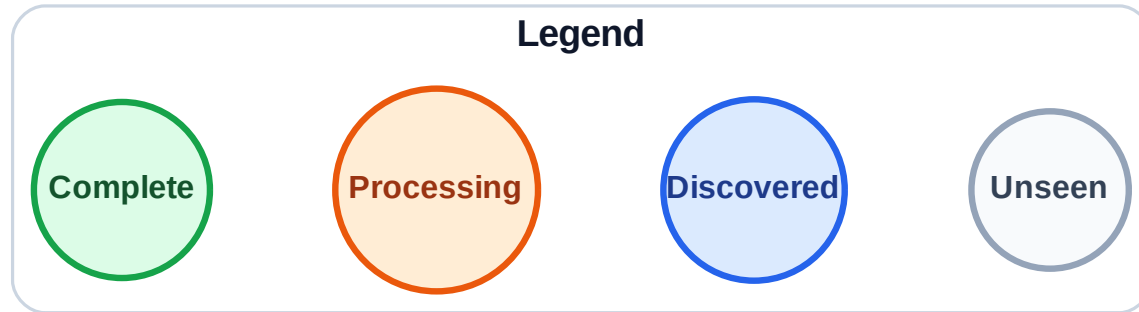
Queue: E → G → H

Visited: A, B, C, F, I, J

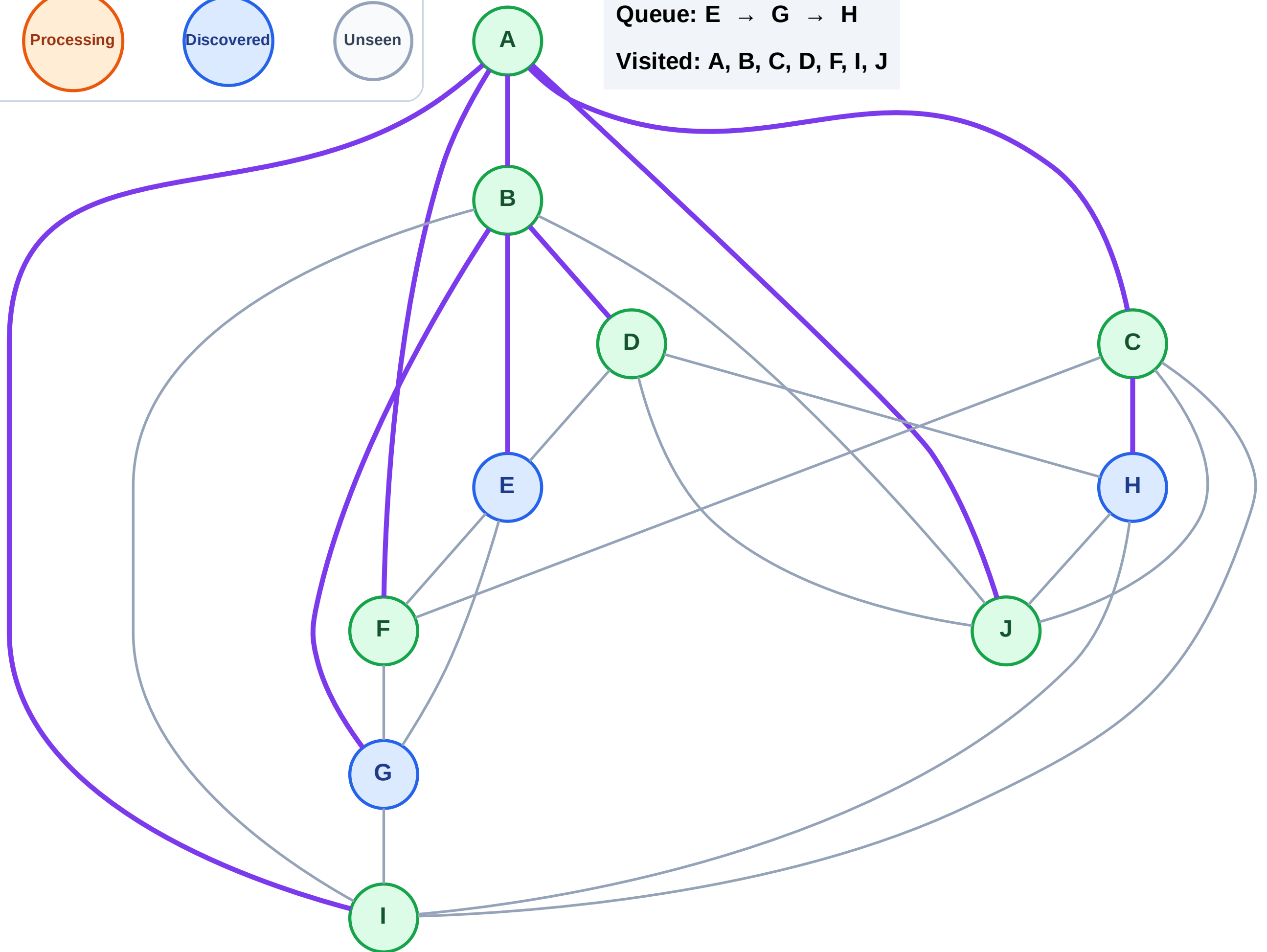


BFS Traversal

Step 15: No new neighbors from D



Queue: E → G → H
Visited: A, B, C, D, F, I, J



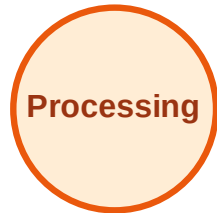
BFS Traversal

Step 16: Process node E

Legend



Complete



Processing



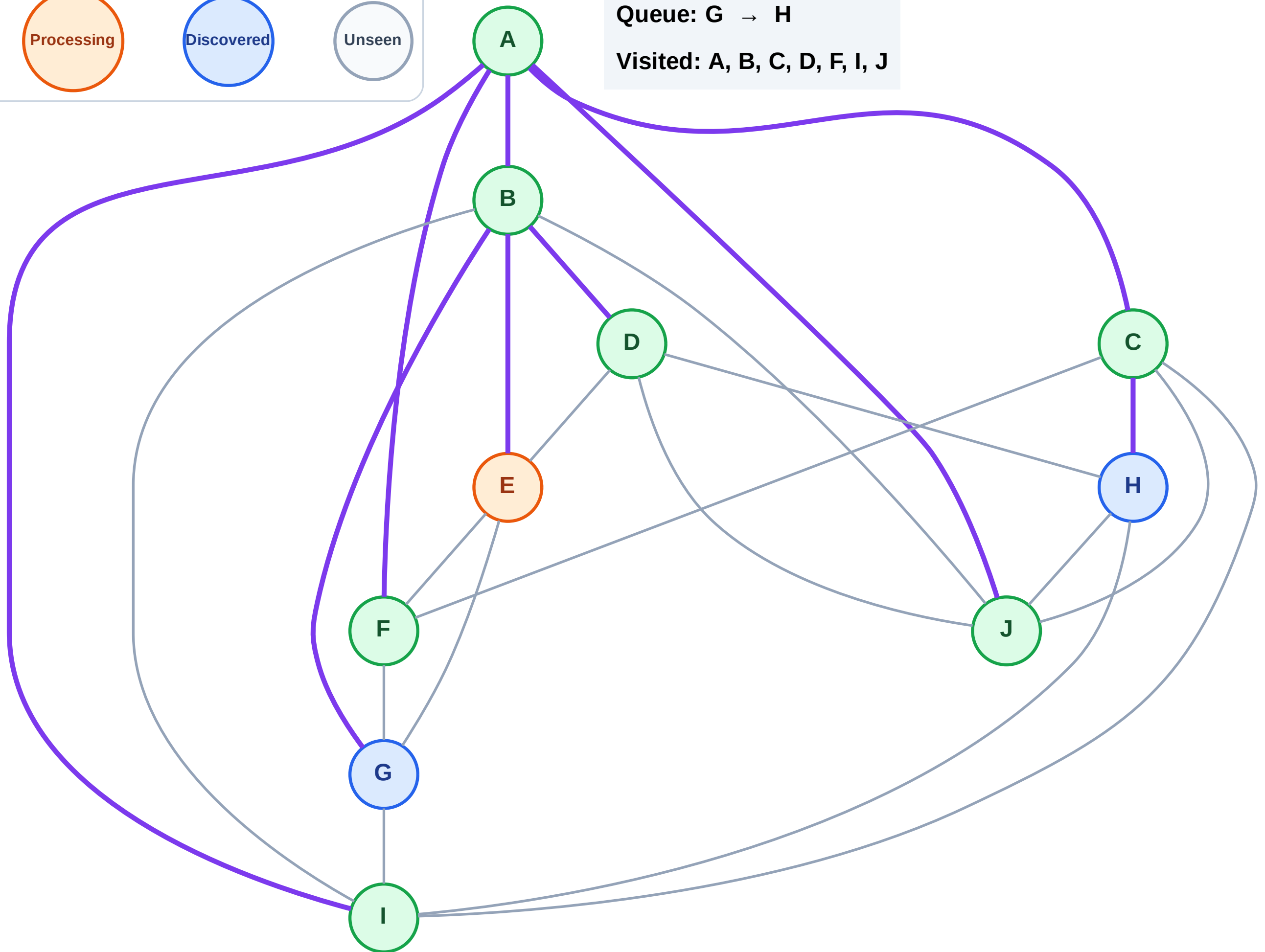
Discovered



Unseen

Queue: G → H

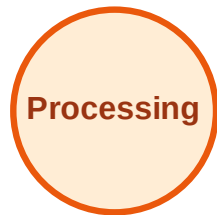
Visited: A, B, C, D, F, I, J



BFS Traversal

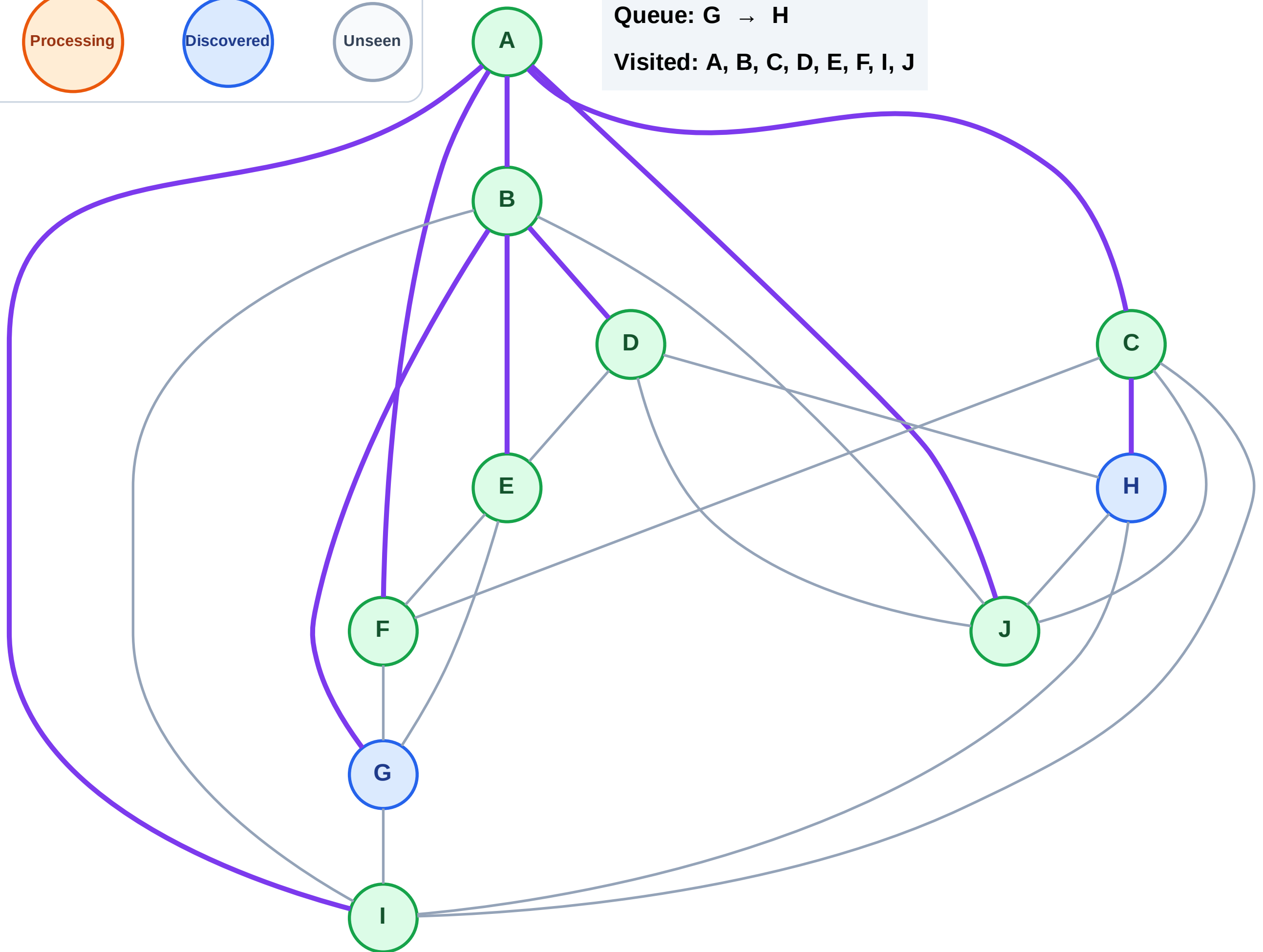
Step 17: No new neighbors from E

Legend



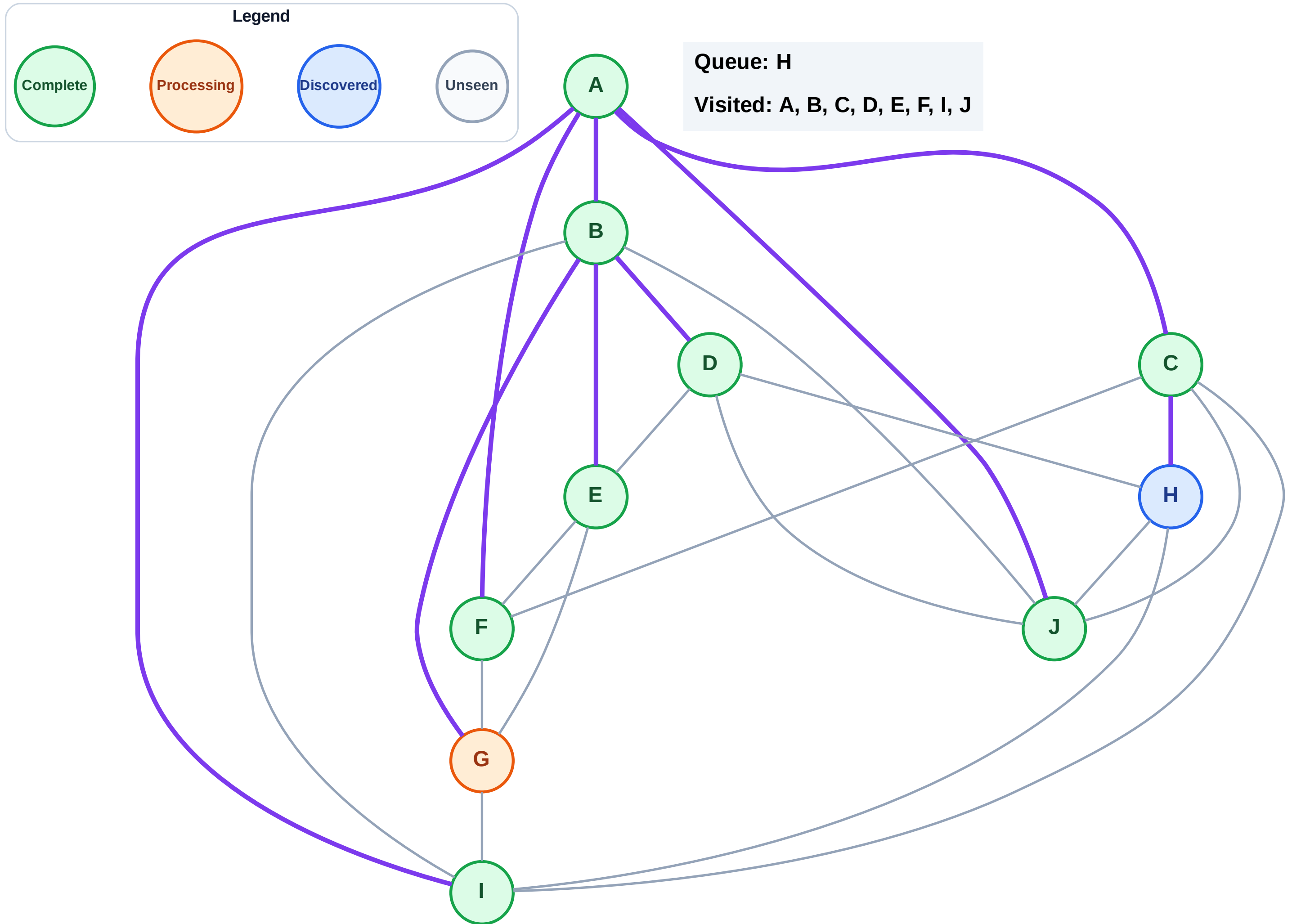
Queue: G → H

Visited: A, B, C, D, E, F, I, J



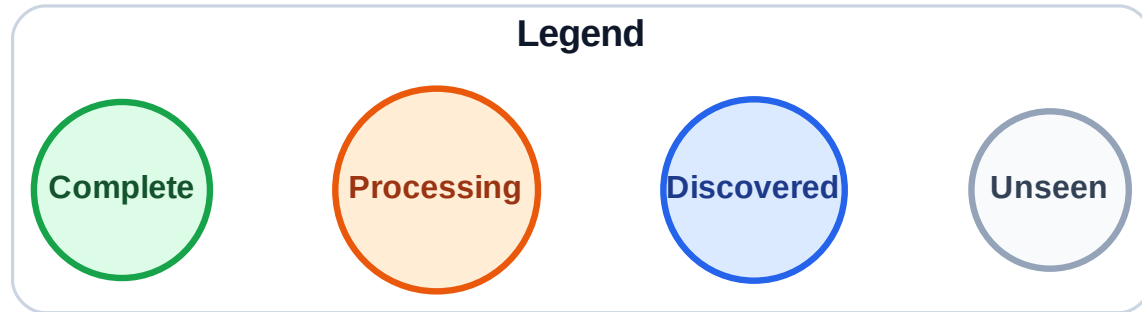
Step 18: Process node G

Step 18: Process node G



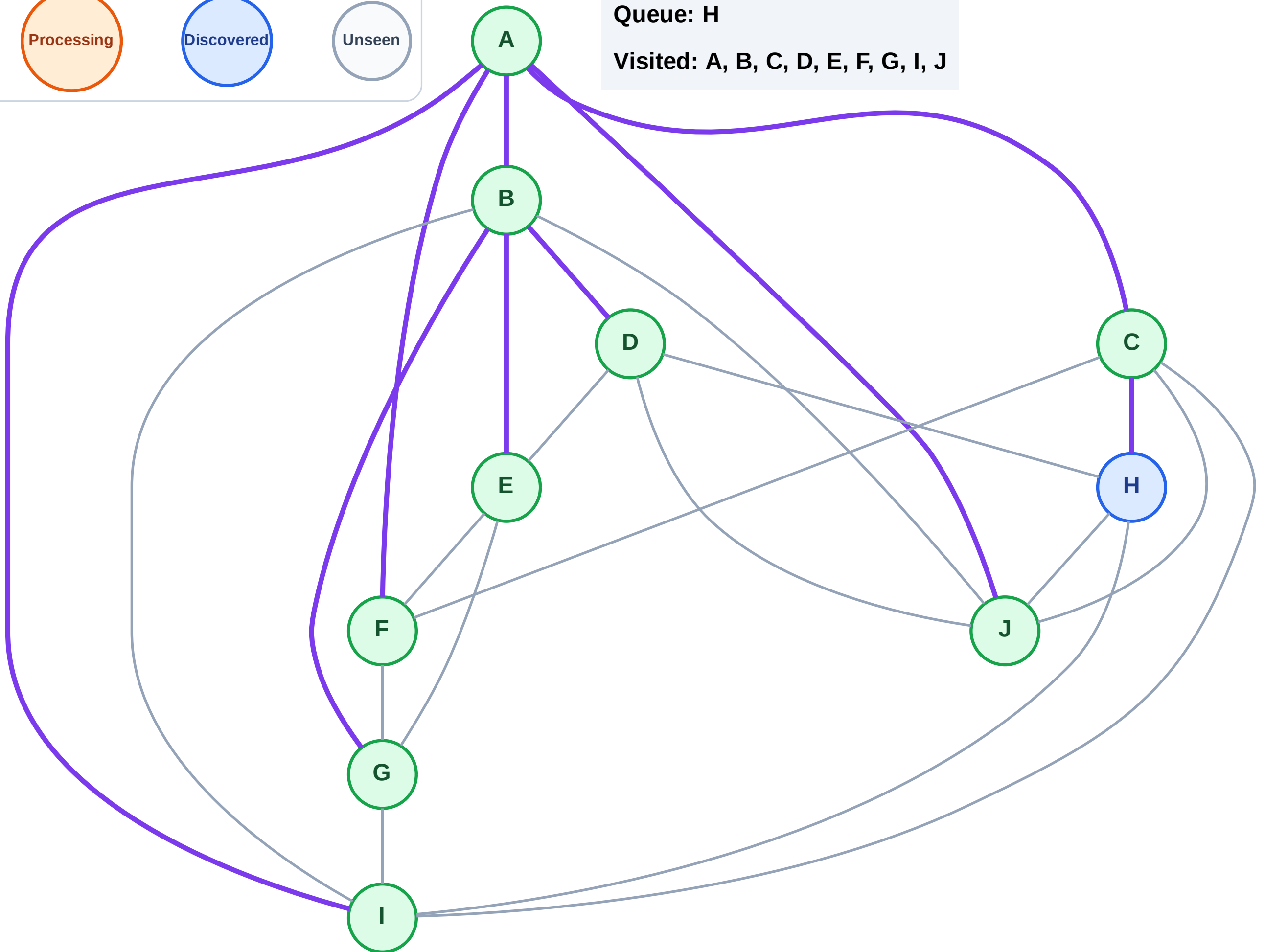
BFS Traversal

Step 19: No new neighbors from G



Queue: H

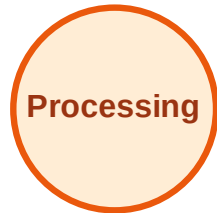
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

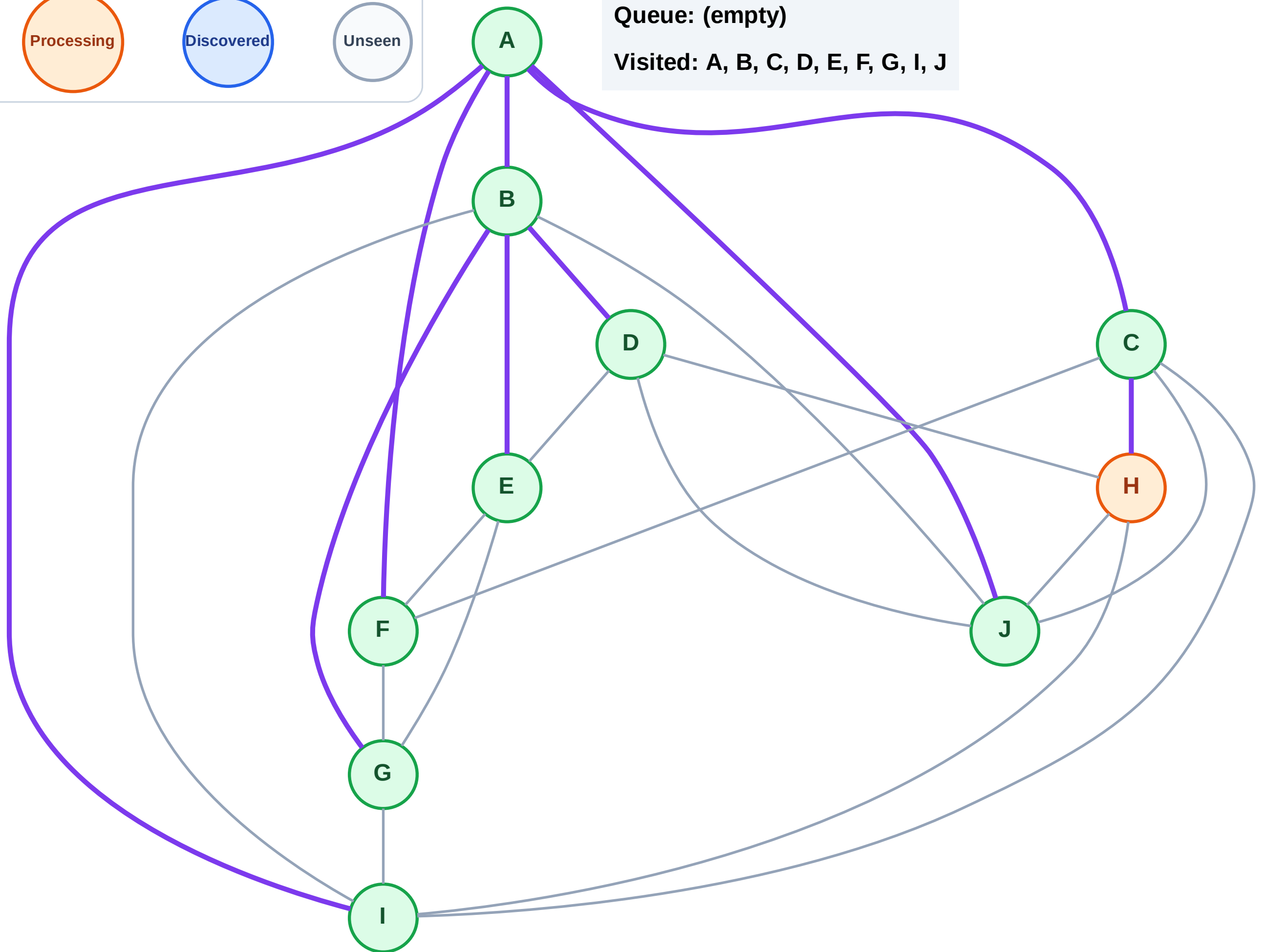
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 21: No new neighbors from H

Legend

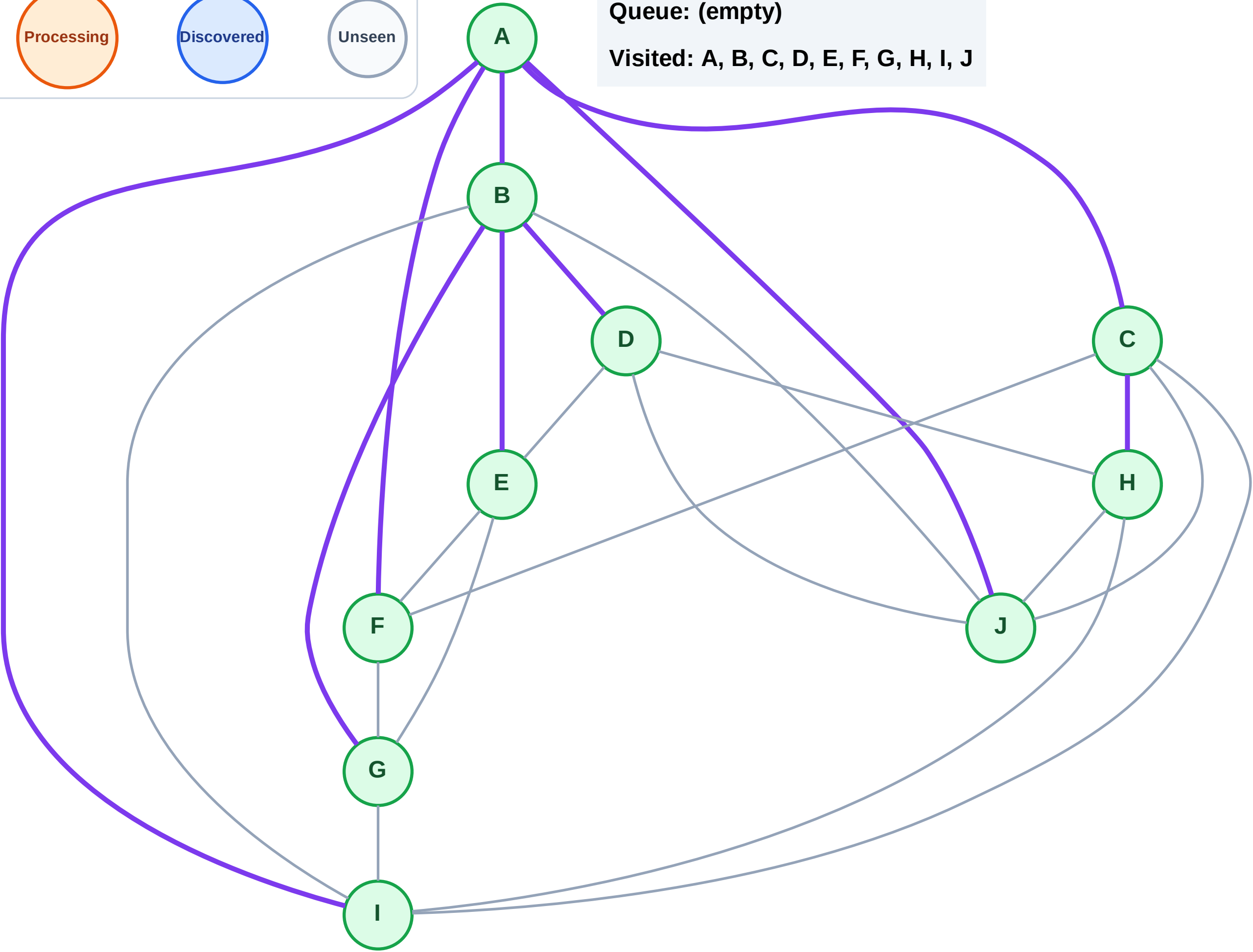
Complete

Processing

Discovered

Unseen

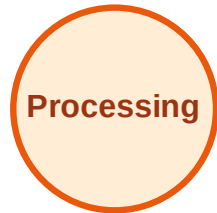
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

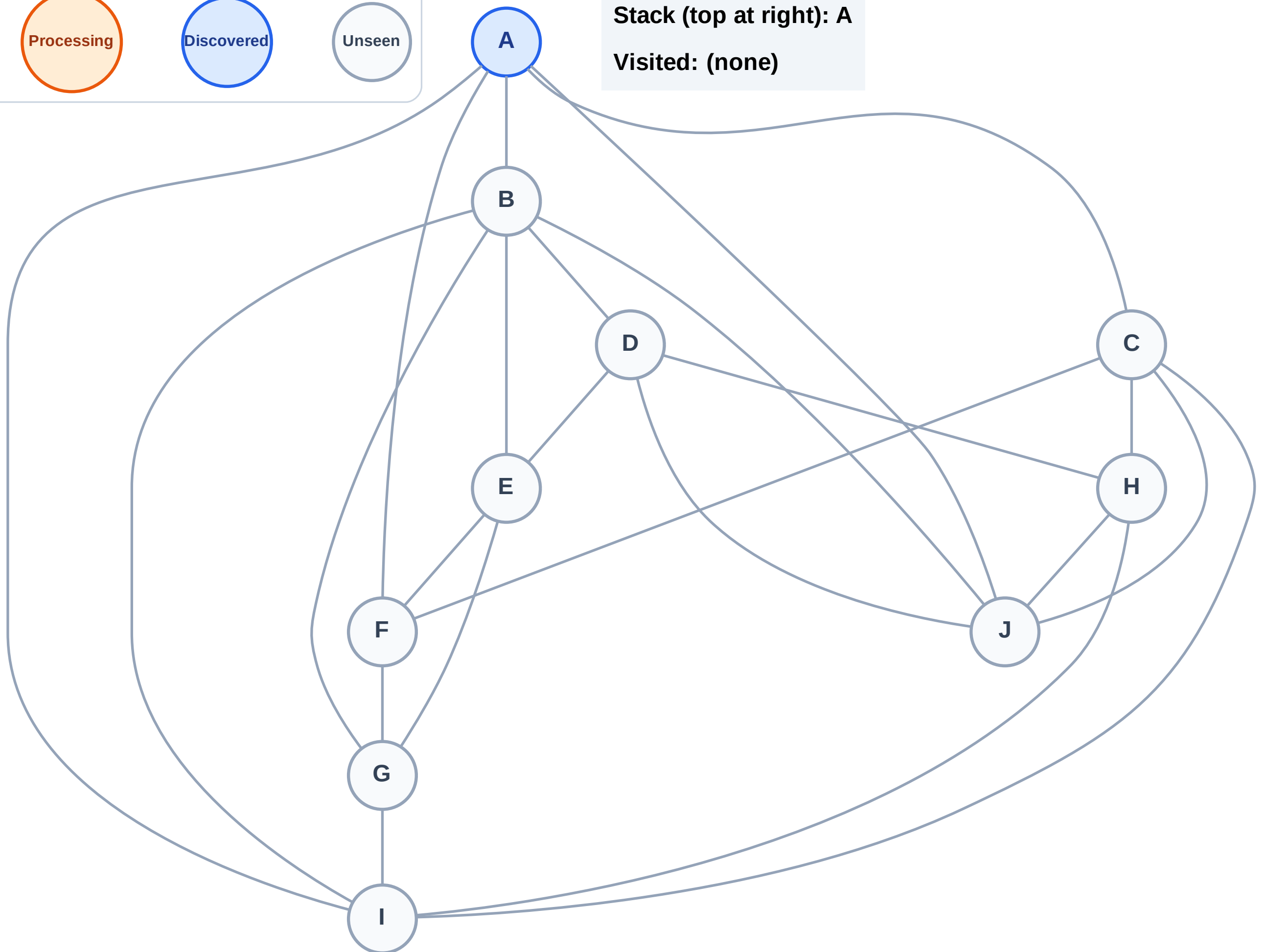
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



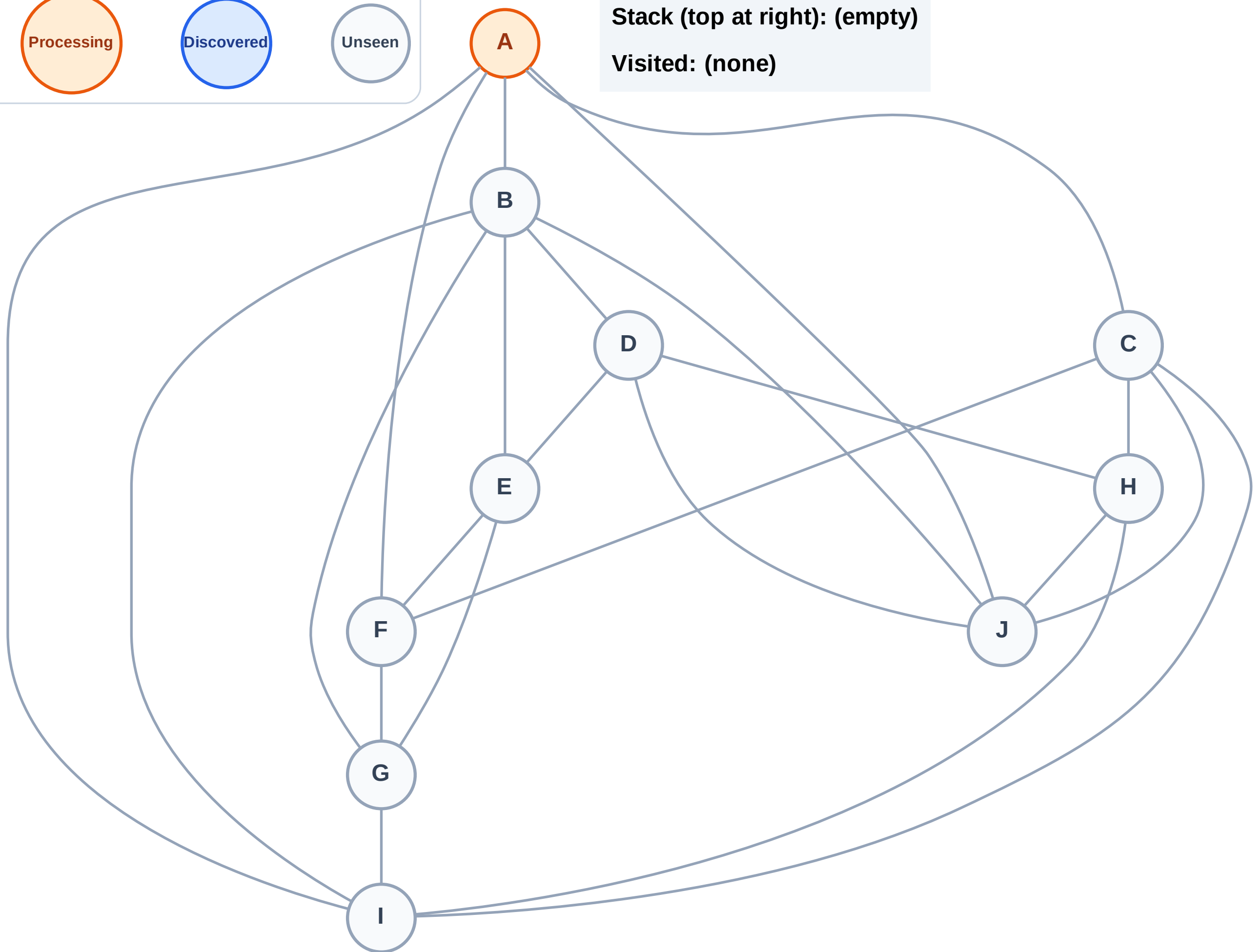
Discovered



Unseen

Stack (top at right): (empty)

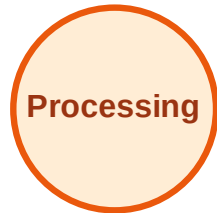
Visited: (none)



DFS Traversal

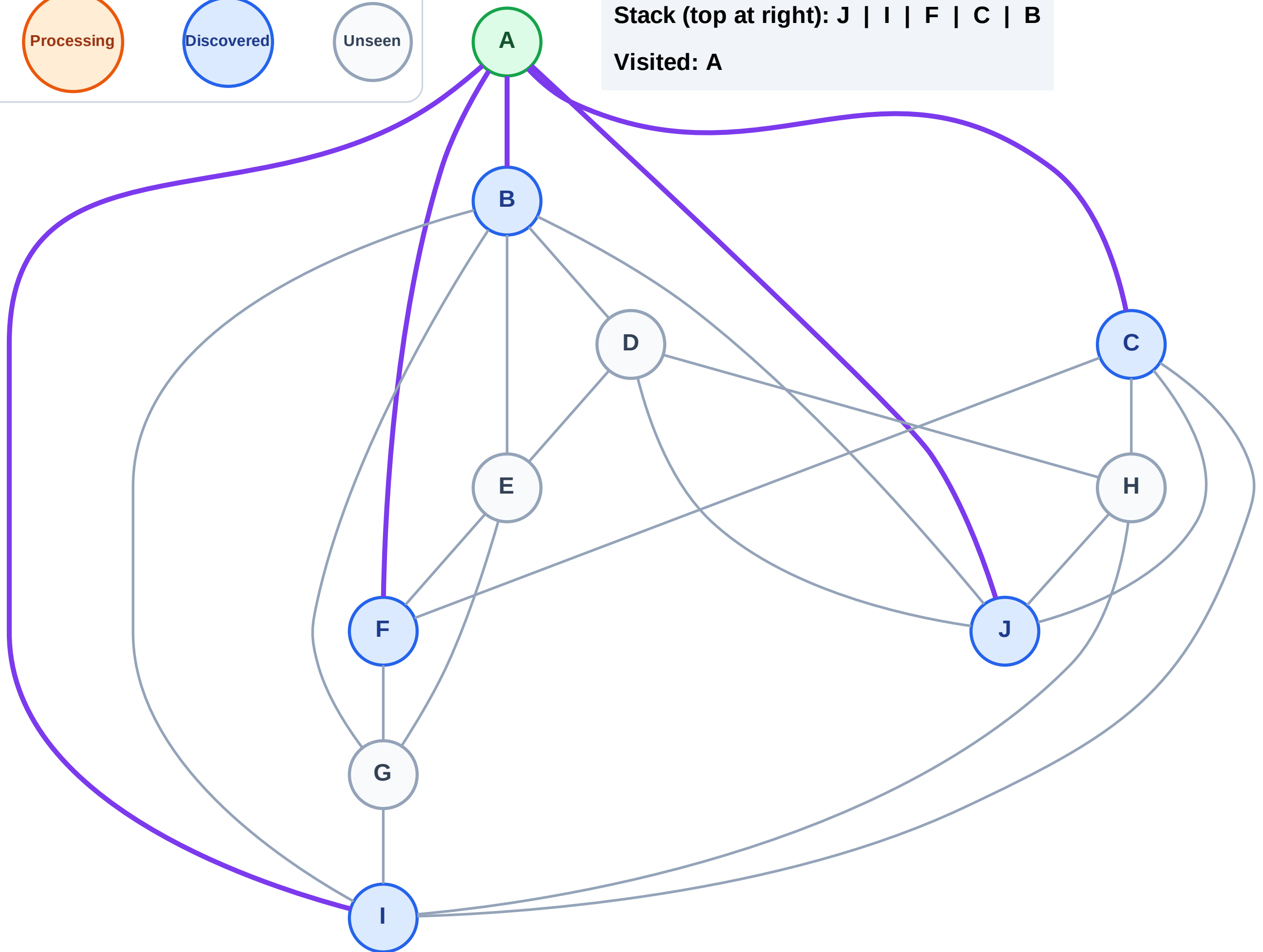
Step 3: Discover J, I, F, C, B from A

Legend



Stack (top at right): J | I | F | C | B

Visited: A



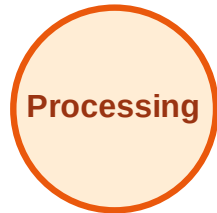
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



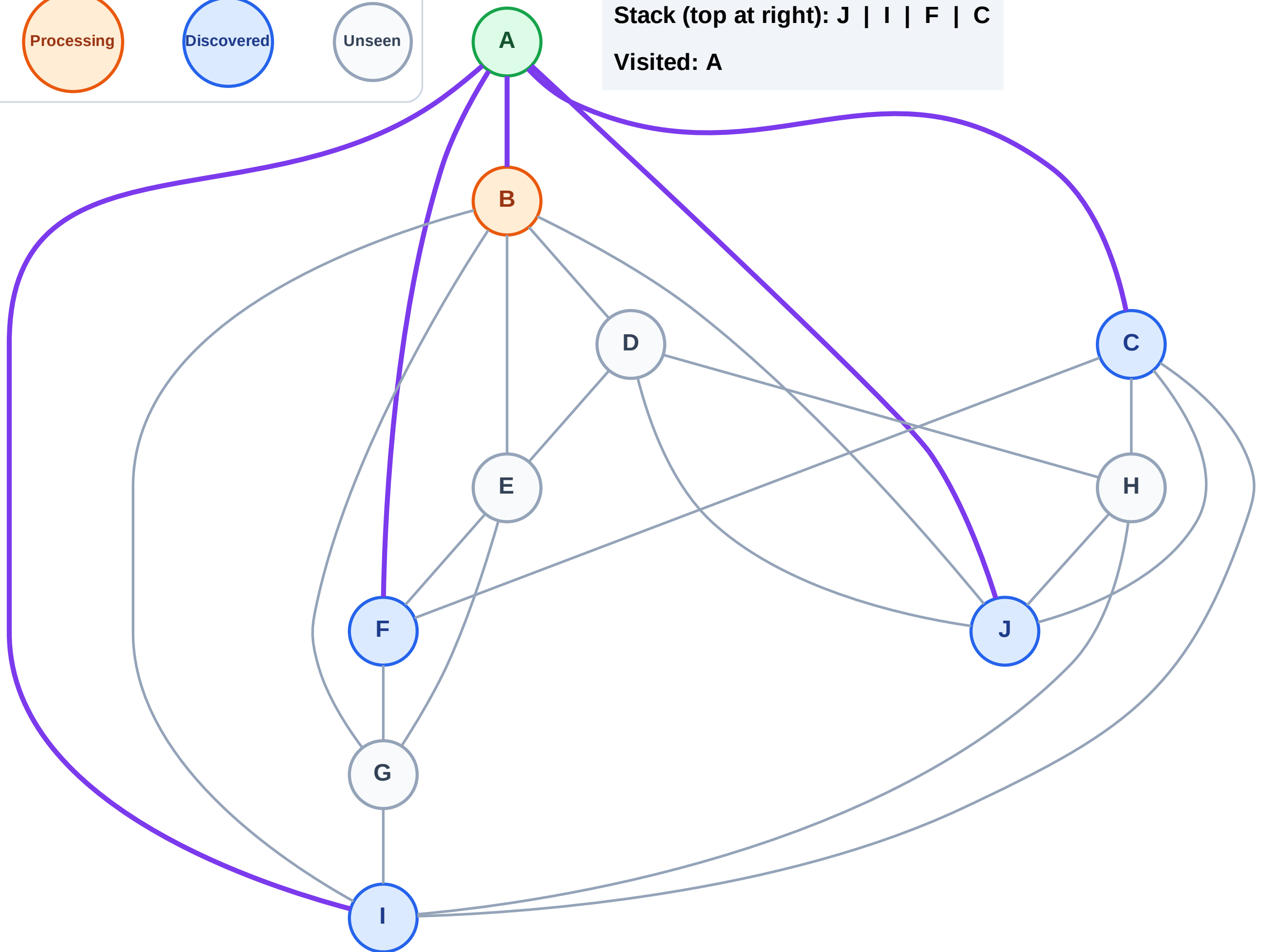
Discovered



Unseen

Stack (top at right): J | I | F | C

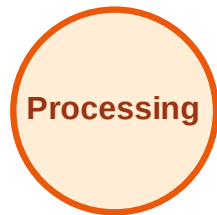
Visited: A



DFS Traversal

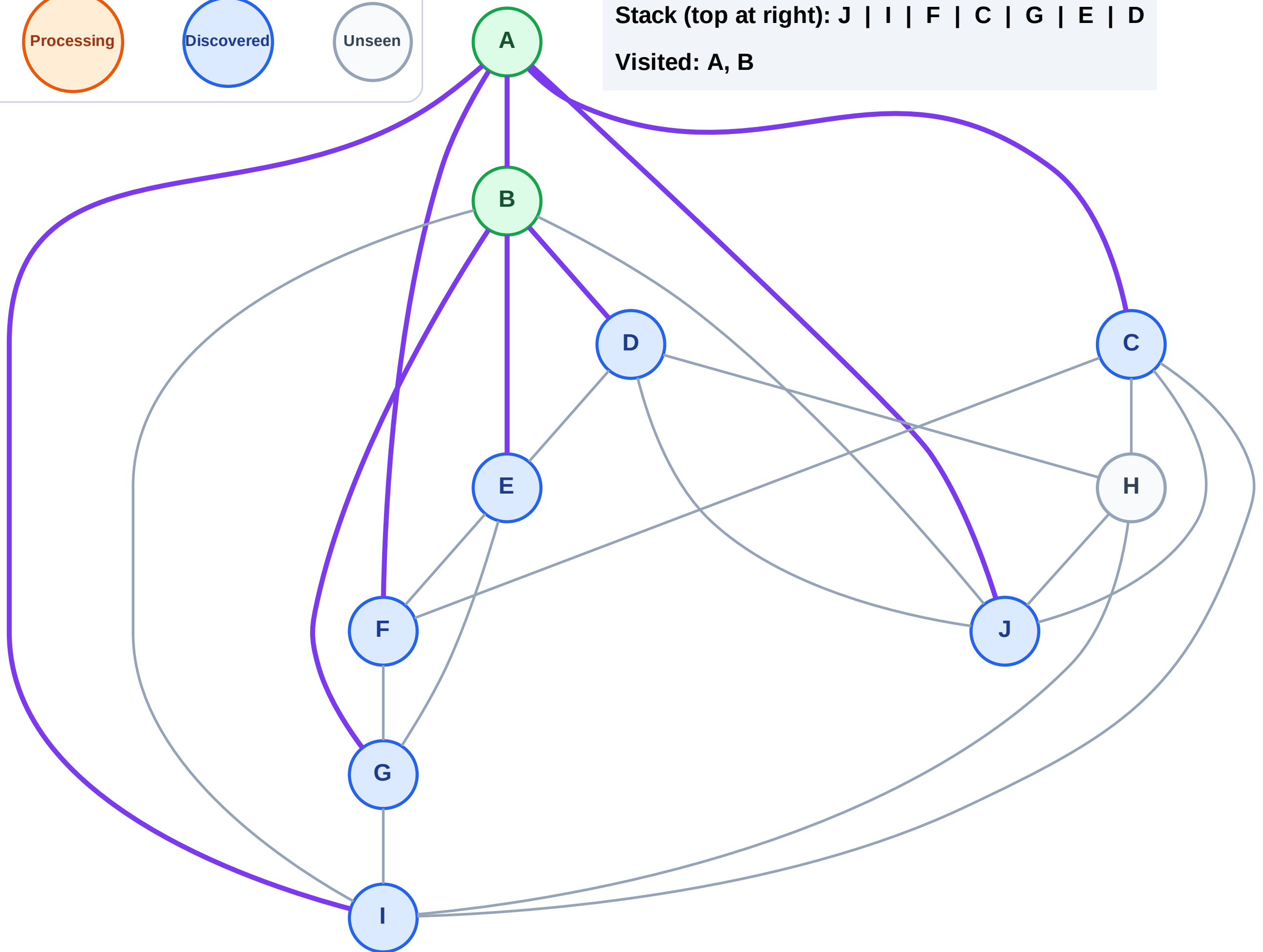
Step 5: Discover G, E, D from B

Legend

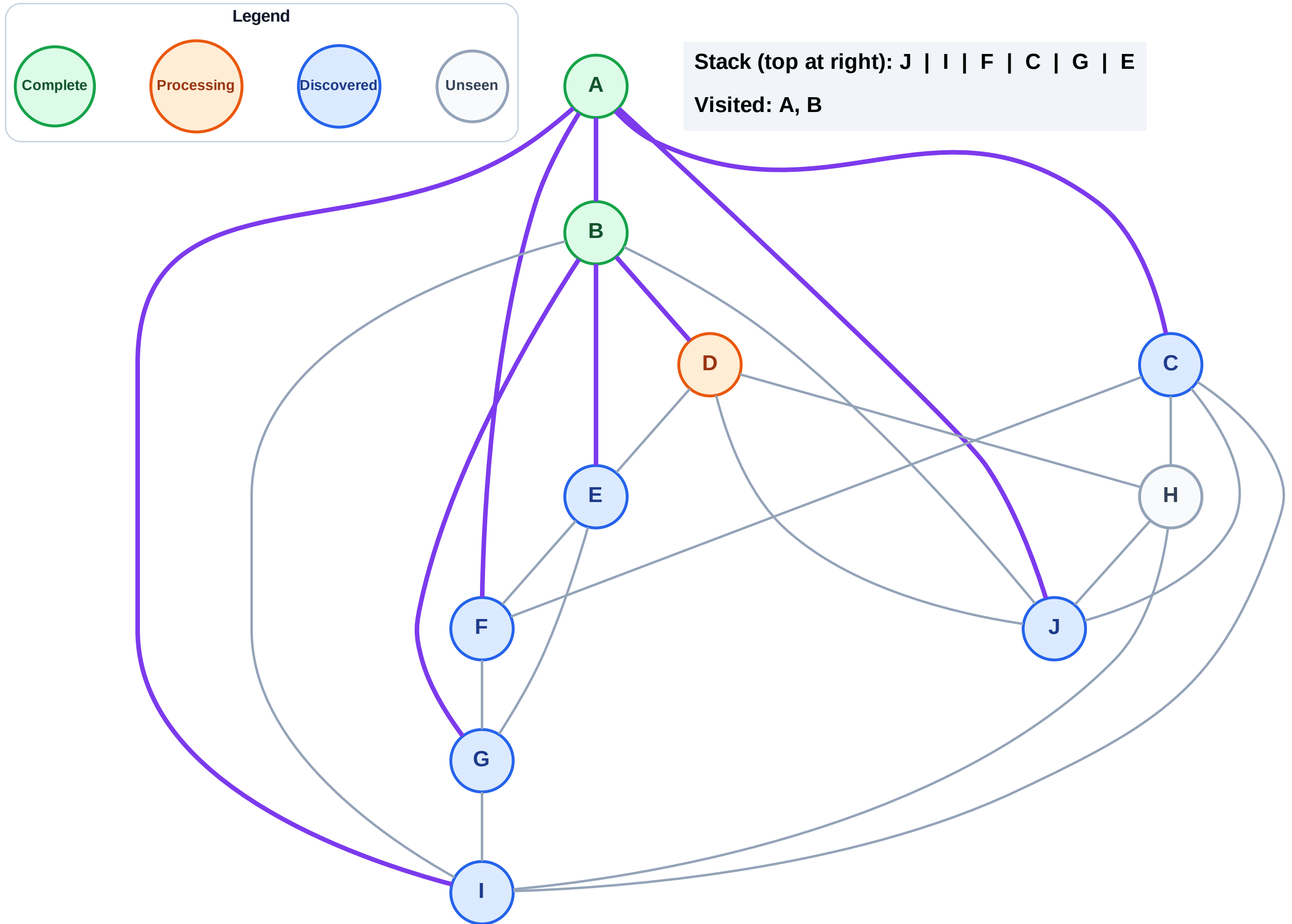


Stack (top at right): J | I | F | C | G | E | D

Visited: A, B



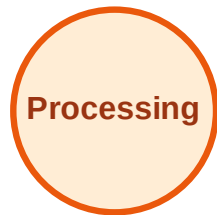
Step 6: Process node D



DFS Traversal

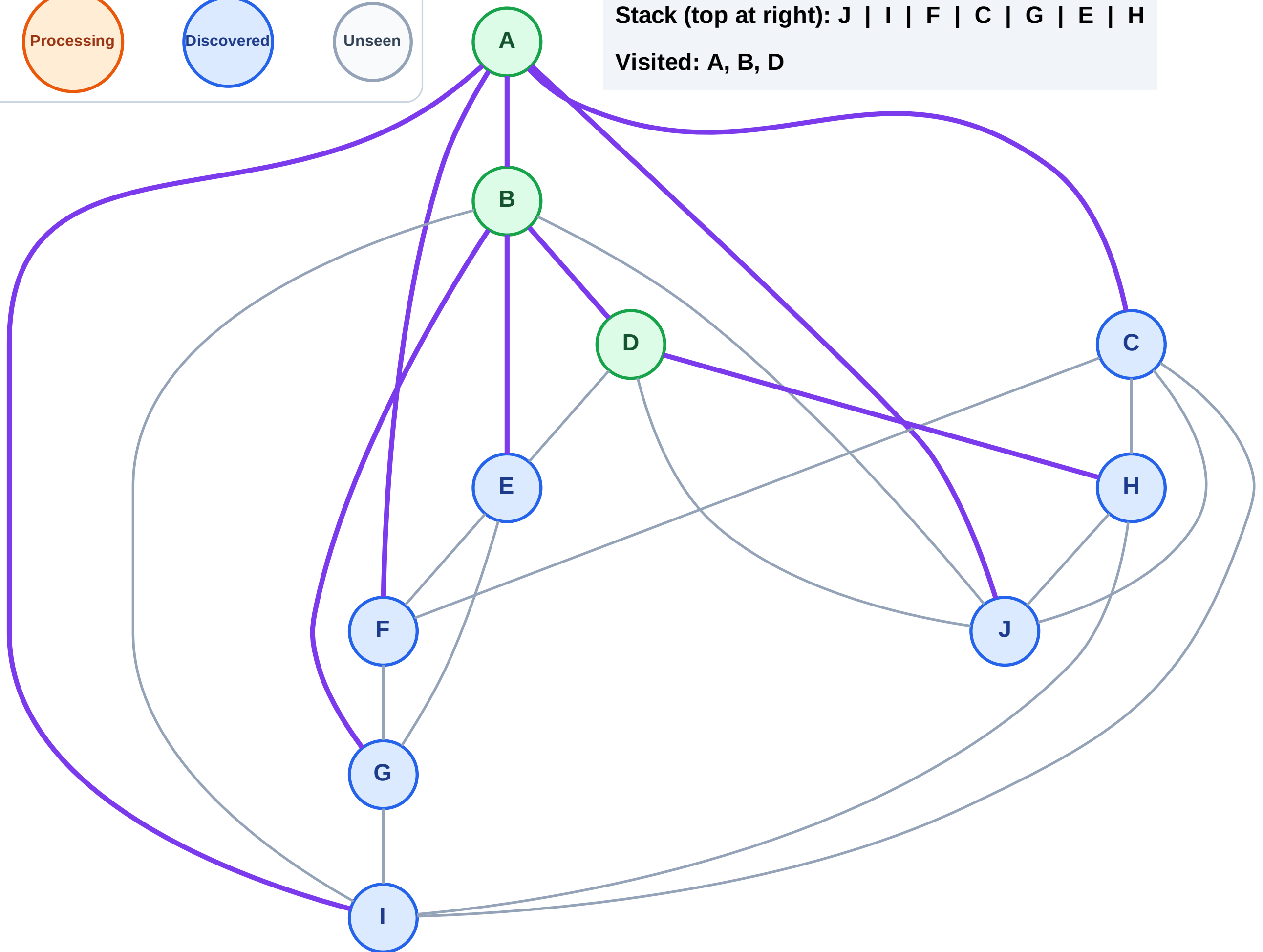
Step 7: Discover H from D

Legend



Stack (top at right): J | I | F | C | G | E | H

Visited: A, B, D



DFS Traversal

Step 8: Process node H

Legend

Complete

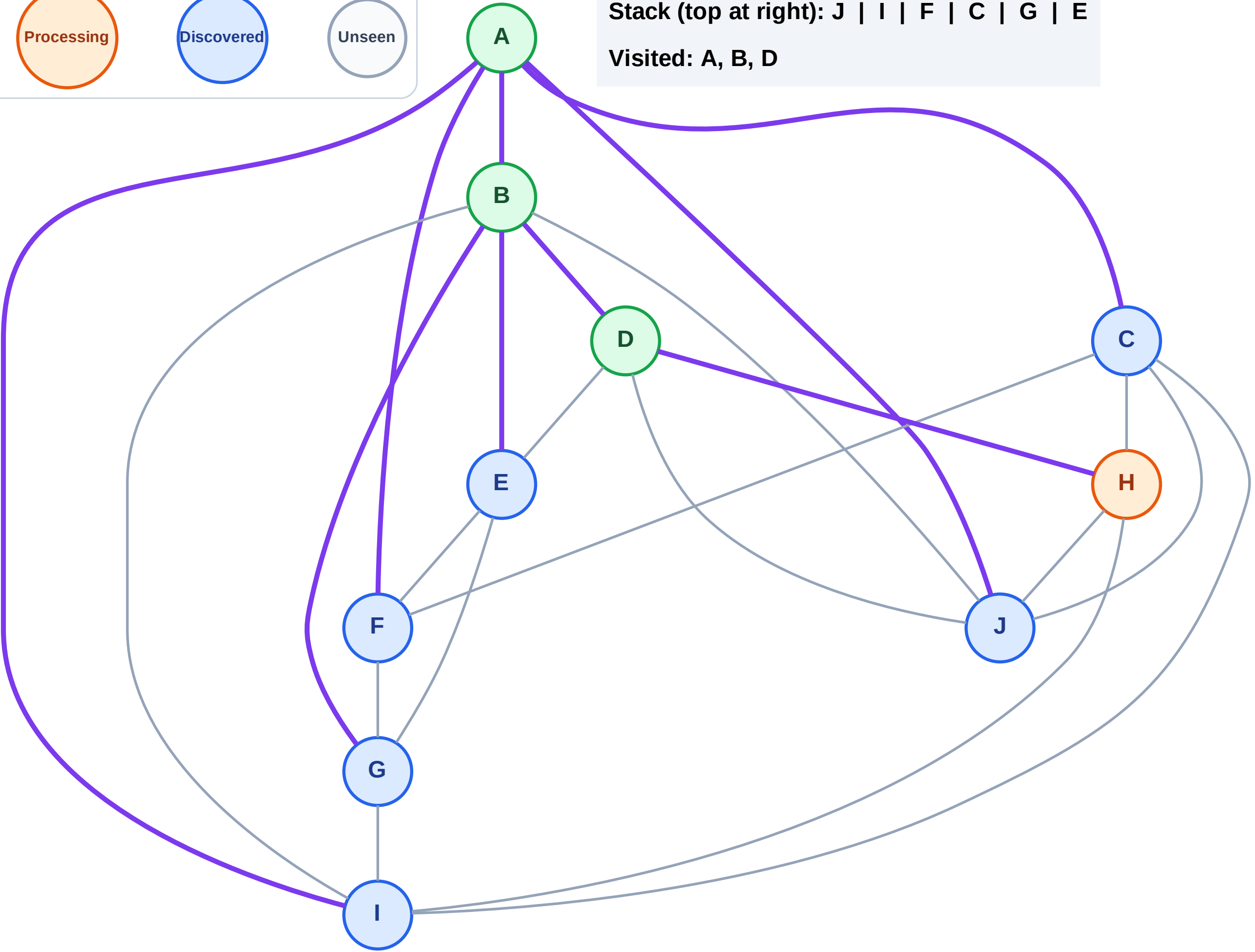
Processing

Discovered

Unseen

Stack (top at right): J | I | F | C | G | E

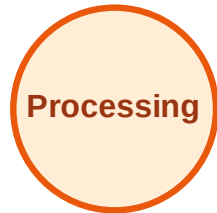
Visited: A, B, D



DFS Traversal

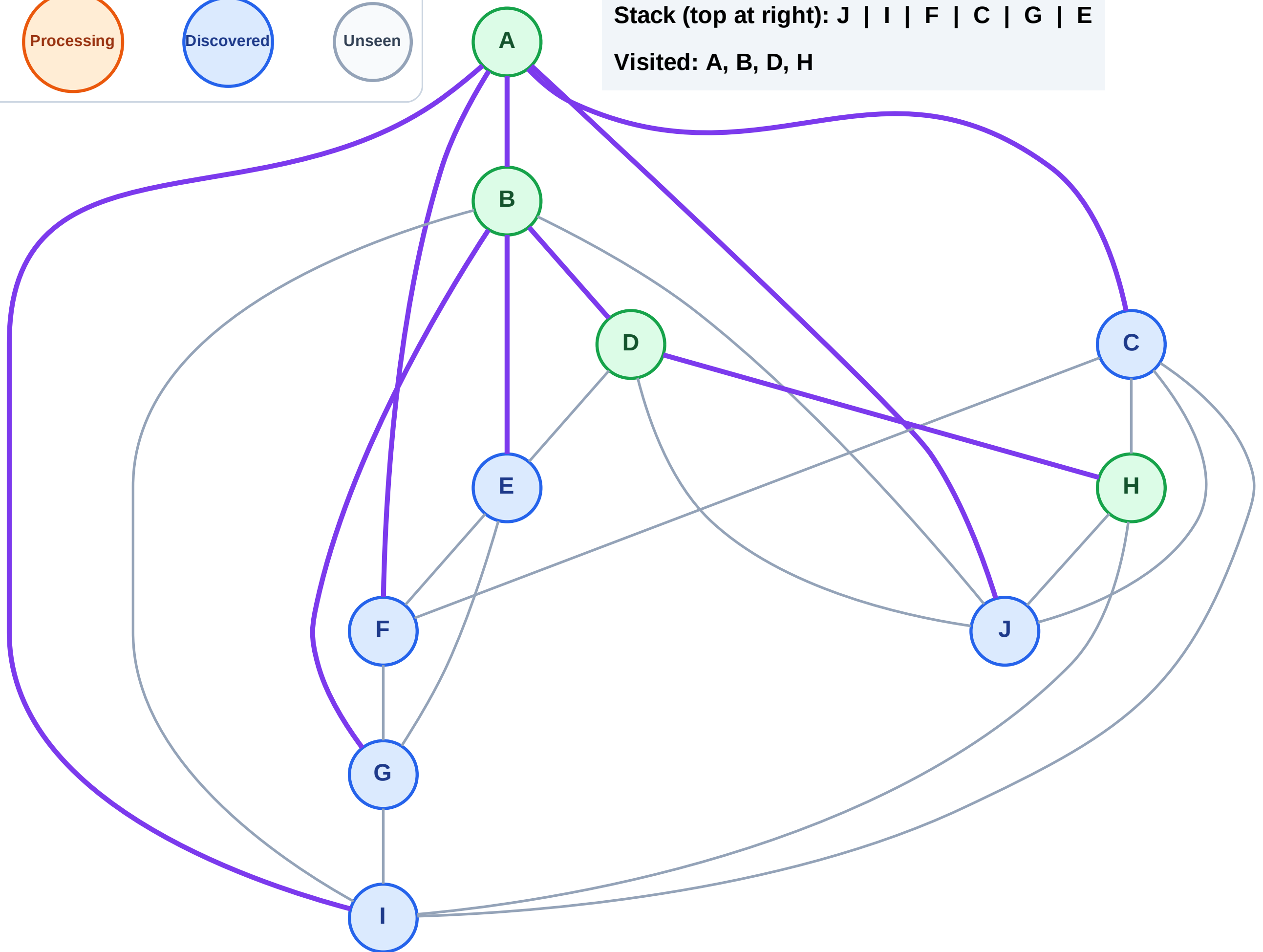
Step 9: No new neighbors from H

Legend



Stack (top at right): J | I | F | C | G | E

Visited: A, B, D, H



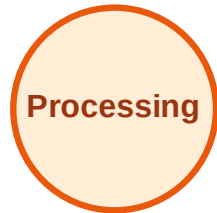
DFS Traversal

Step 10: Process node E

Legend



Complete



Processing



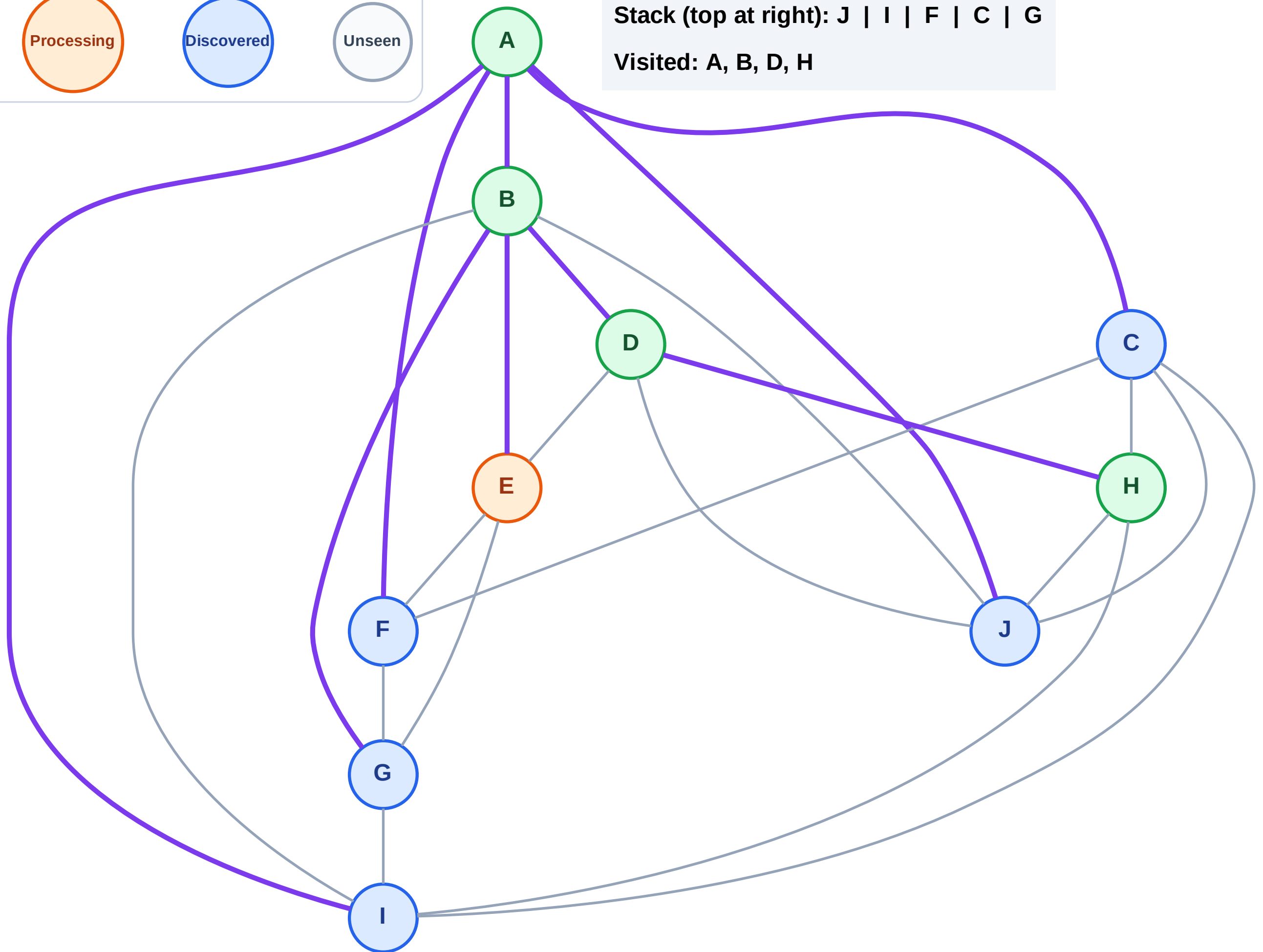
Discovered



Unseen

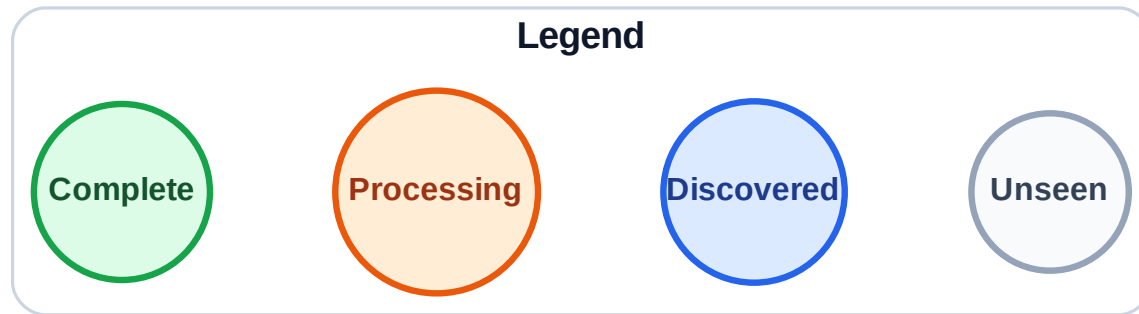
Stack (top at right): J | I | F | C | G

Visited: A, B, D, H



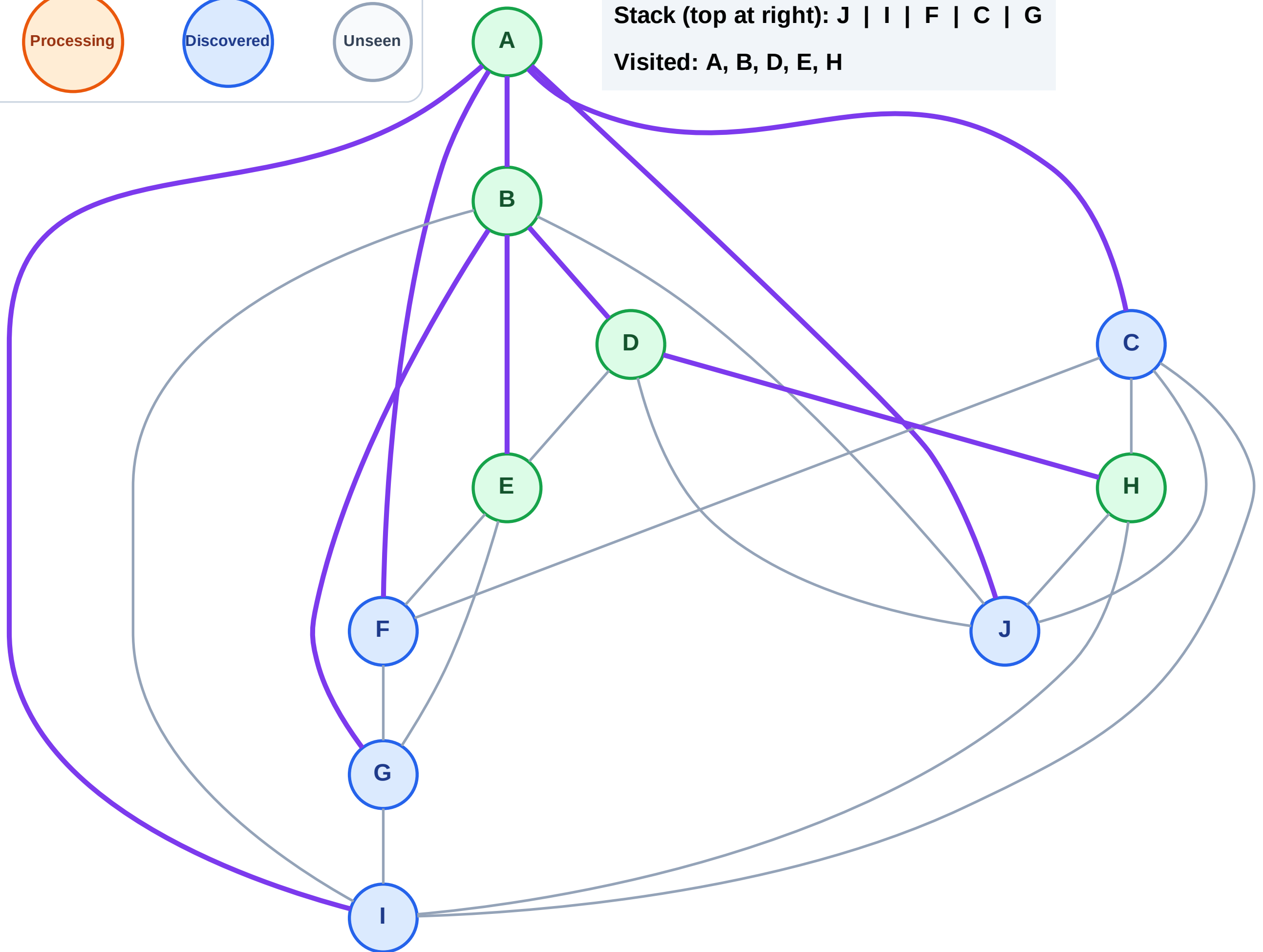
DFS Traversal

Step 11: No new neighbors from E



Stack (top at right): J | I | F | C | G

Visited: A, B, D, E, H



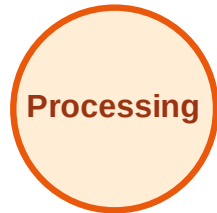
DFS Traversal

Step 12: Process node G

Legend



Complete



Processing



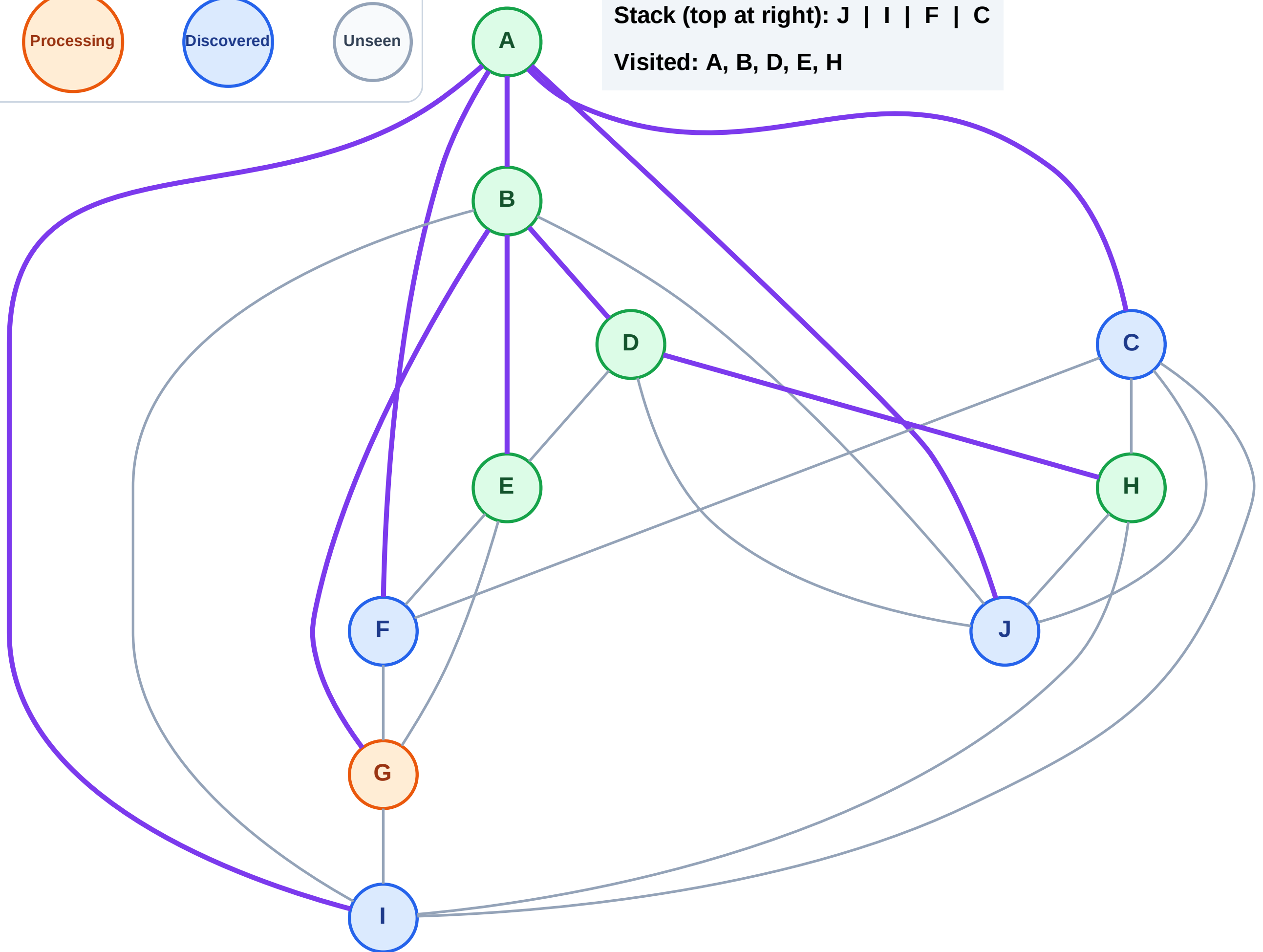
Discovered



Unseen

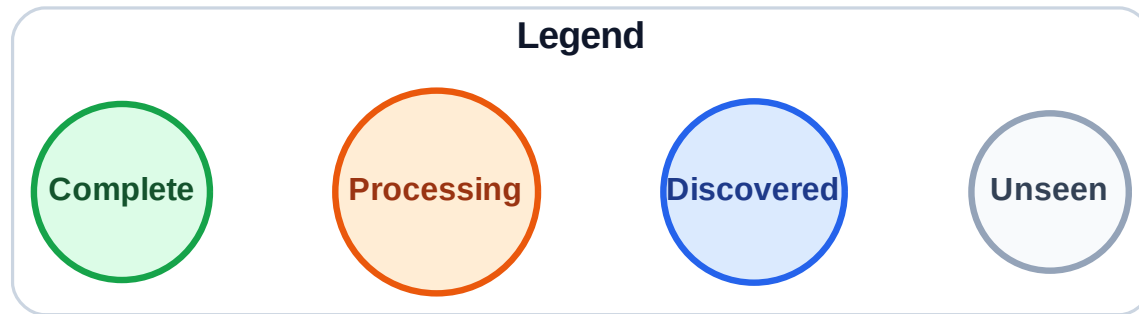
Stack (top at right): J | I | F | C

Visited: A, B, D, E, H



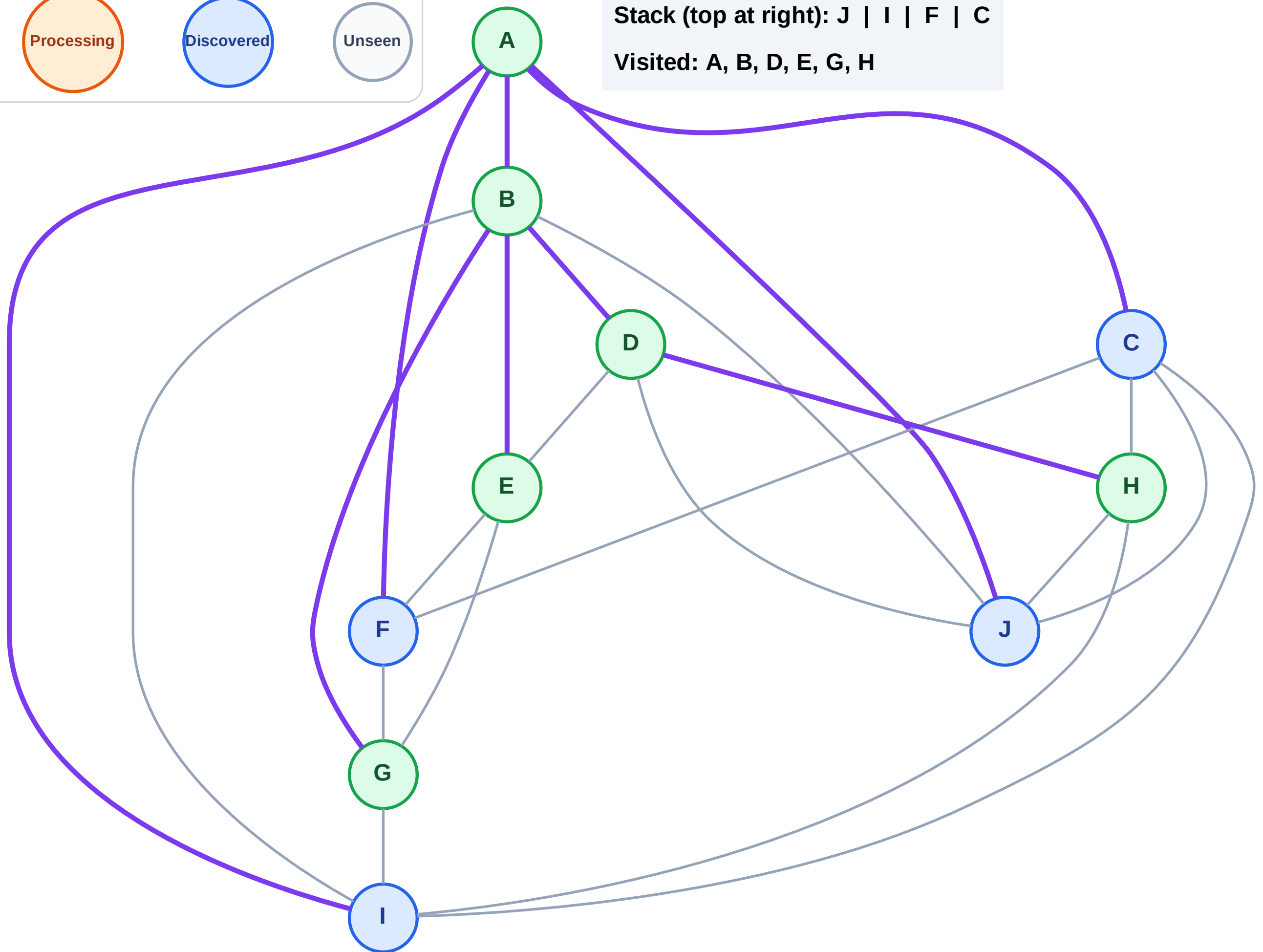
DFS Traversal

Step 13: No new neighbors from G

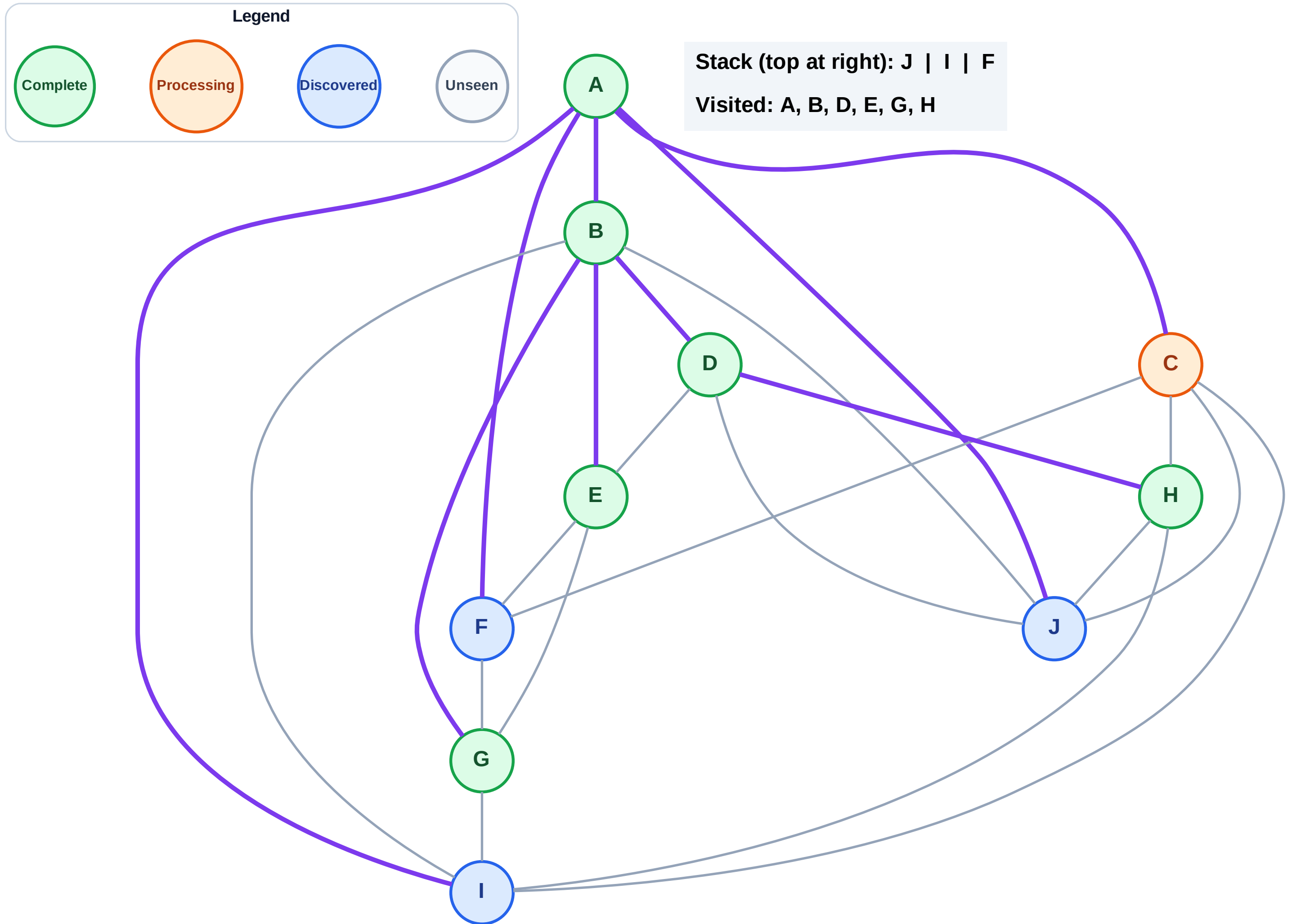


Stack (top at right): J | I | F | C

Visited: A, B, D, E, G, H



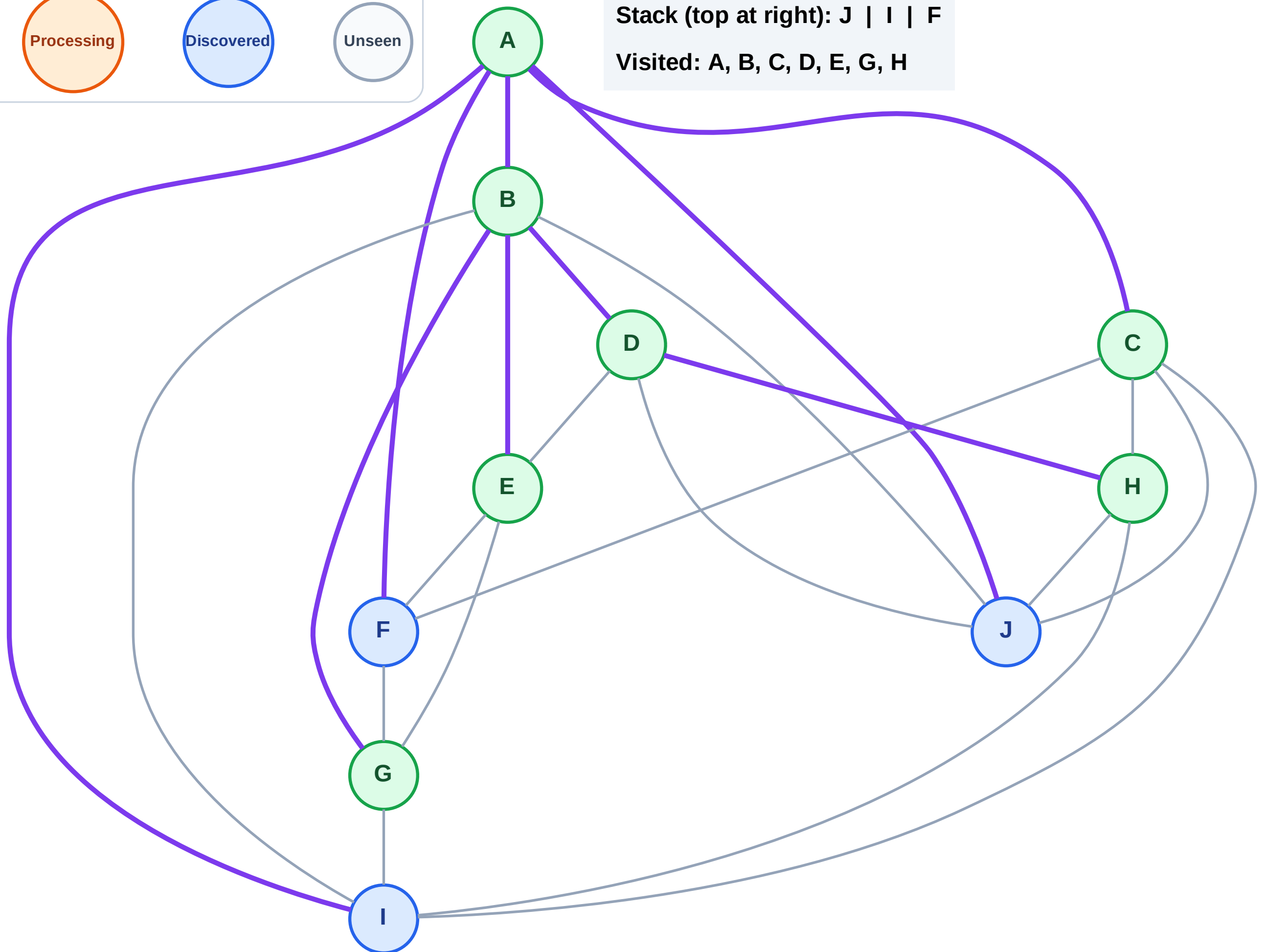
Step 14: Process node C



Step 15: No new neighbors from C

Stack (top at right): J | I | F

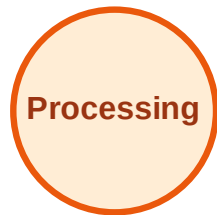
Visited: A, B, C, D, E, G, H



DFS Traversal

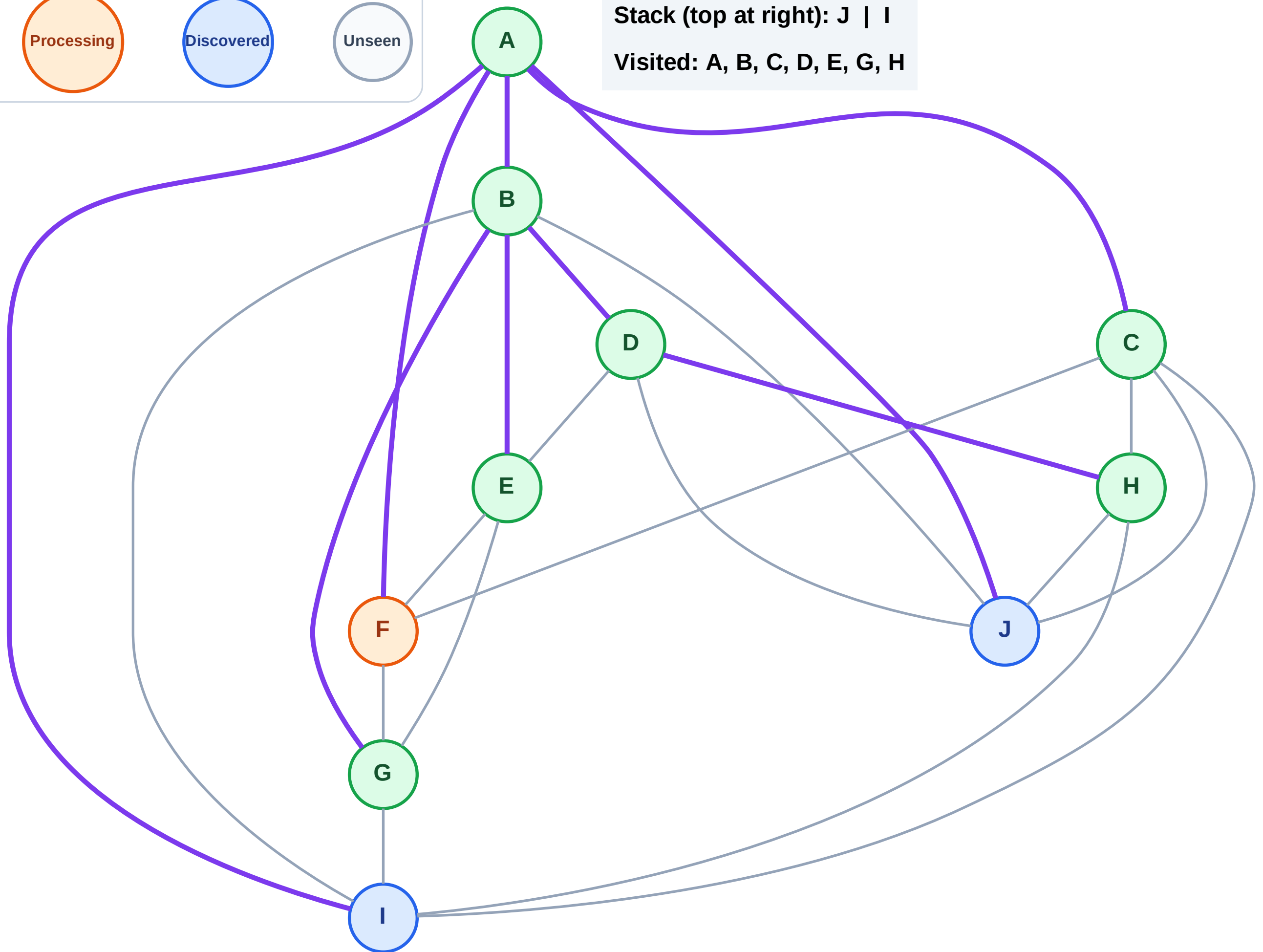
Step 16: Process node F

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, G, H



DFS Traversal

Step 17: No new neighbors from F

Legend

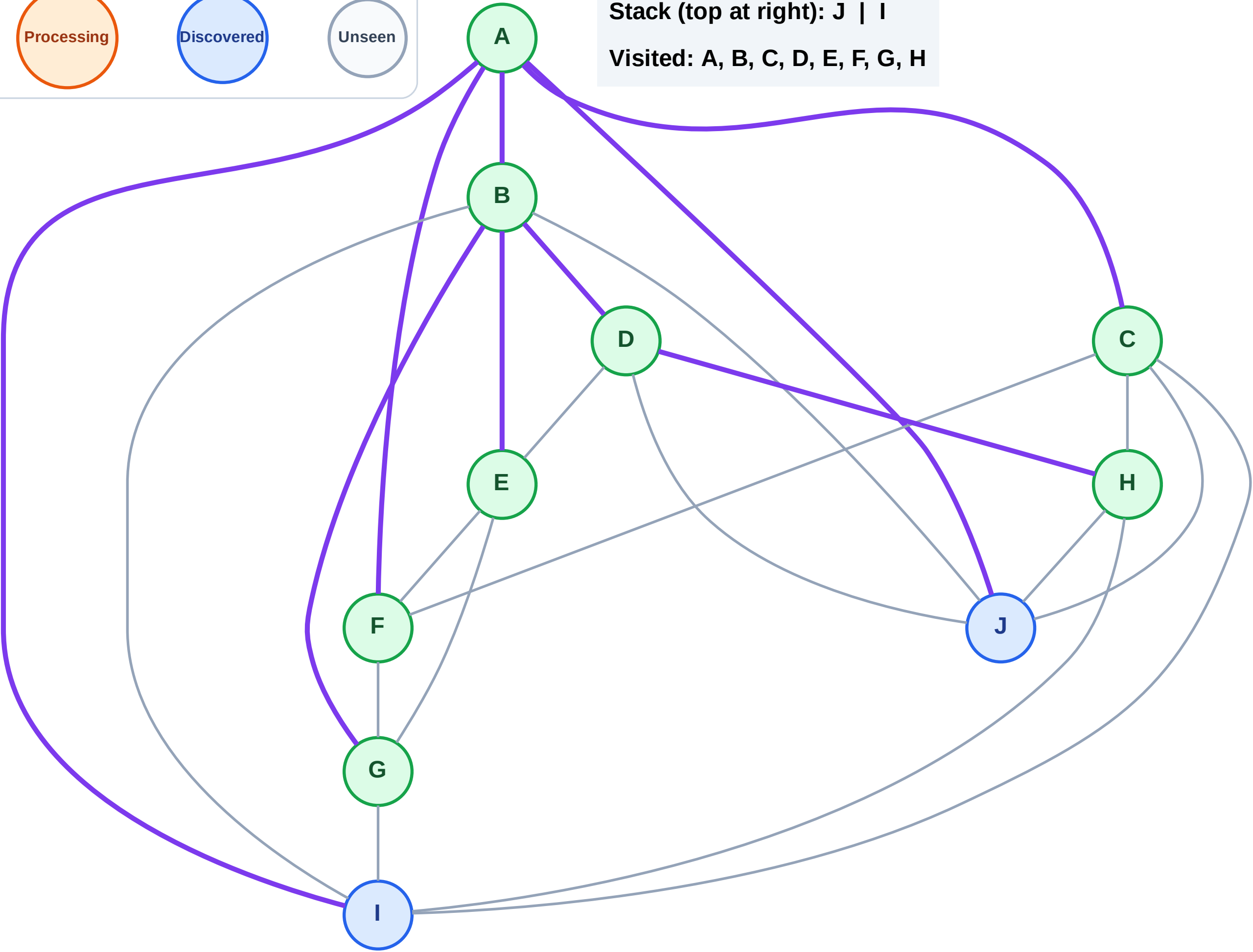
Complete

Processing

Discovered

Unseen

Stack (top at right): J | I
Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node I

Legend

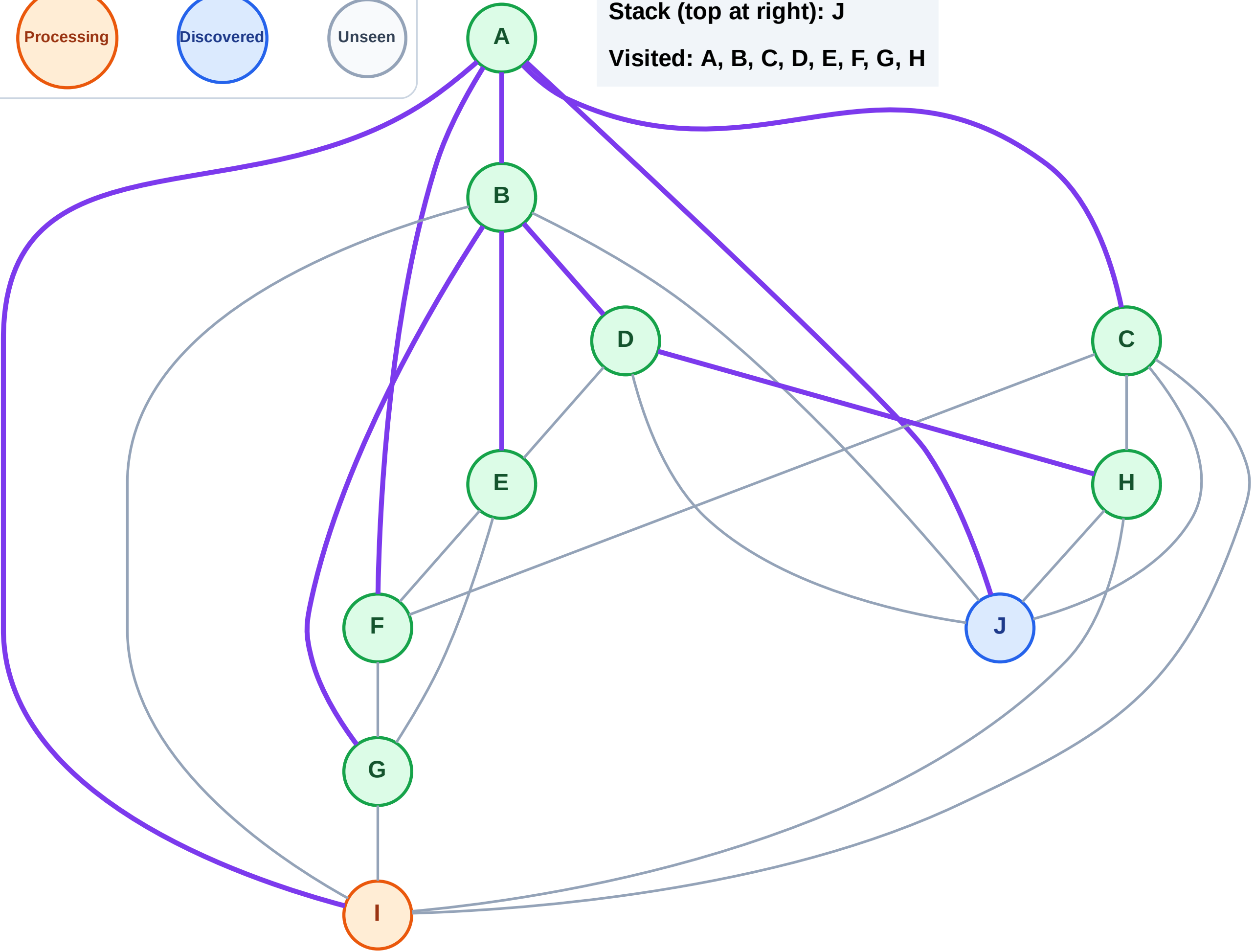
Complete

Processing

Discovered

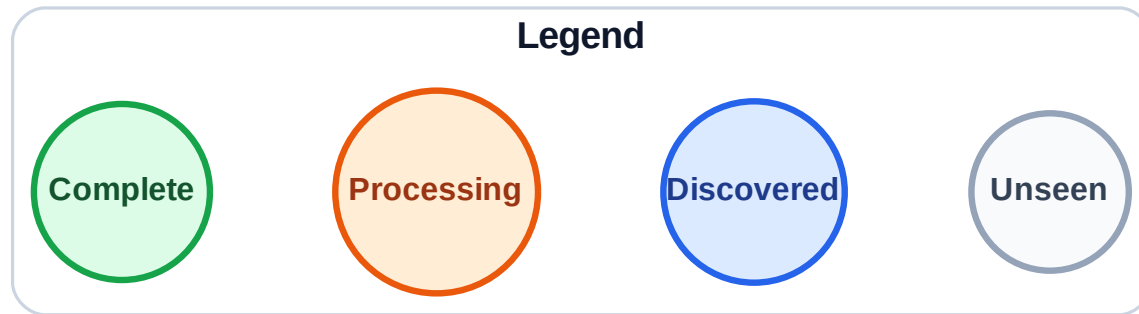
Unseen

Stack (top at right): J
Visited: A, B, C, D, E, F, G, H

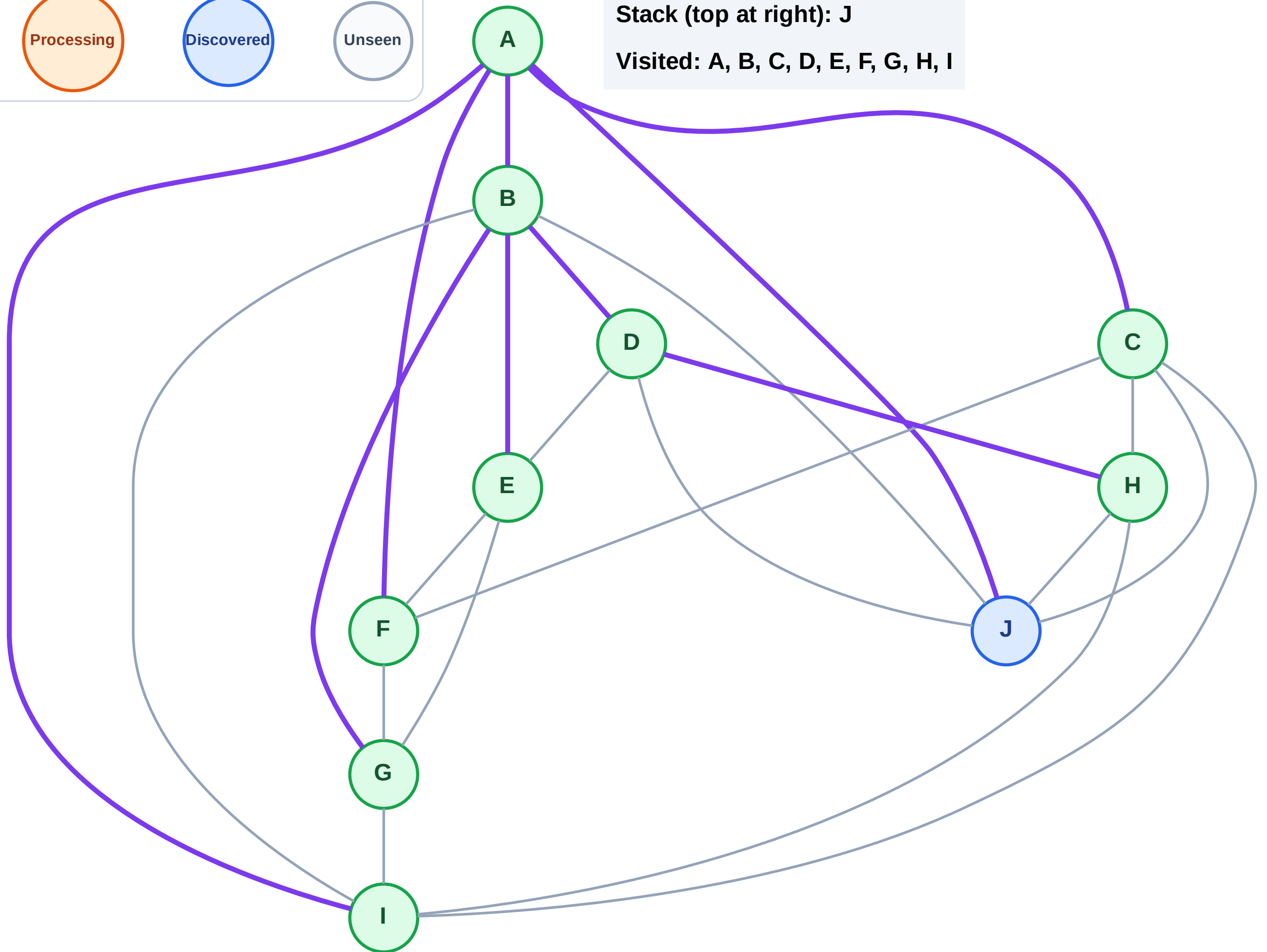


DFS Traversal

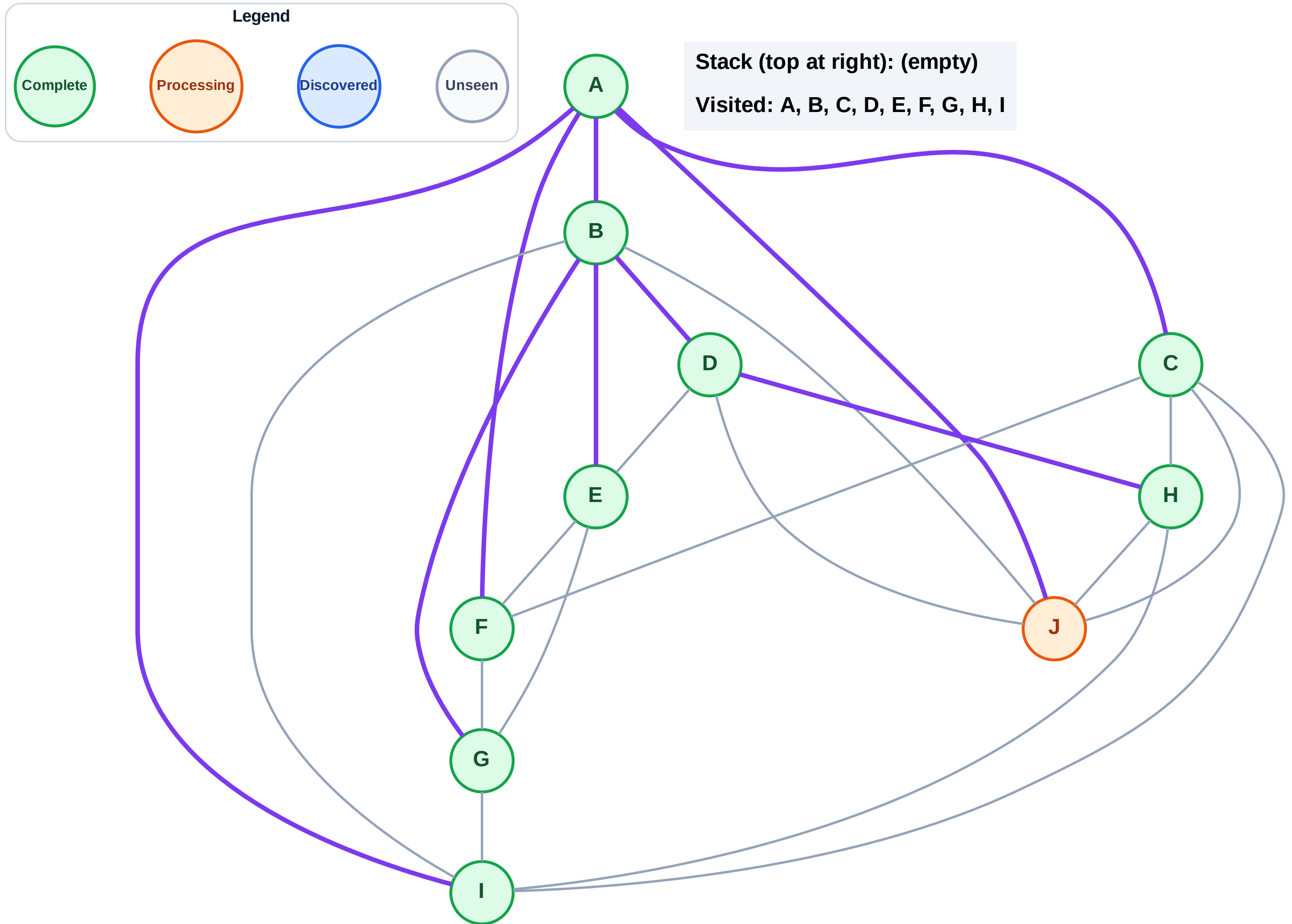
Step 19: No new neighbors from I



Stack (top at right): J
Visited: A, B, C, D, E, F, G, H, I



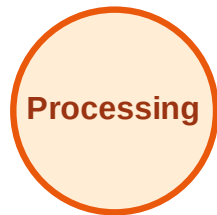
Step 20: Process node J



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

